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Improvement of the Three-Dimensional Inspection Tooling

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Abstract

In the aeronautical sector, maintaining dimensional accuracy during part inspection is critical for ensuring product quality, particularly as components become lighter and more flexible. Fixtures are increasingly used to limit deformation during measurements, but many current solutions are not well adapted to parts with high geometric complexity or low stiffness. This research was conducted in collaboration with the LAUAK Group to address a specific industrial challenge: the inability of a dimensional self-supporting inspection tool to maintain the required flatness tolerance of a sealing flange support used on the LEAP-B1 engine for the Boeing 737 MAX. The observed deformations (up to 0.15 mm) significantly exceeded the acceptable limit of 0.025 mm, despite the current system's use of flexible spring supports.

The aim of this study was to redesign the inspection fixture using a structured engineering approach in order to improve its mechanical behavior and ensure the part's dimensional conformity. The project followed the DMAIC (Define, Measure, Analyse, Improve, Control) methodology, combining CAD modeling in CATIA, mechanical simulations in ANSYS, and analytical calculations to evaluate and optimize various reinforcement strategies. Different hypotheses were tested, including the addition of ribs, extra thickness, and internal ring stiffeners. Each configuration was assessed for deformation reduction and industrial feasibility. The most effective designs achieved a reduction in vertical displacement by over 80% while remaining within the acceptable mass and manufacturability constraints.

The results demonstrate that structural reinforcement of the fixture enables significant improvement in measurement accuracy for flexible aeronautical parts. This research highlights the importance of combining numerical simulations with design-for-manufacture principles in metrology tooling. Although no physical prototype could be manufactured within the project timeline, the proposed solutions offer a validated digital alternative and form a solid foundation for future industrial implementation. More broadly, the methodology developed in this study can be extended to other flexible fixturing challenges in aeronautical manufacturing environments, contributing to enhanced conformity assurance and reduced reliance on client concessions.

Keywords: dimensional control, fixture design, finite element analysis, aeronautical metrology, deformation, DMAIC

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ChatGPT was used to correct typos in code, refine LaTeX formatting, and enhance the clarity of the English grammar and syntax of the report. Additionally, DeepL provided translation support, ensuring that the content was both accurate and well-articulated.

As a non-native English speaker, I also referred to the book "Science Research Writing for Non-Native Speakers of English" by Glasman-Deal H. [1] to improve the technical quality and clarity of my scientific writing.

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1 Introduction

1.1 Industrial and Technical Context

Ensuring the dimensional conformity of critical components has become a strategic challenge for modern aerospace manufacturing, particularly in the context of increasingly digitalised quality control.

The aerospace industry is facing growing demands in terms of safety, quality, and dimensional precision. The geometric conformity of components is a critical issue, directly affecting the performance, reliability, and regulatory compliance of structural assemblies. As geometries become more complex and tolerances tighter, control strategies and associated devices must evolve to ensure measurement reliability.

Coordinate Measuring Machines (CMMs) play a key role in guaranteeing dimensional accuracy and geometric compliance of aeronautical parts, thereby contributing directly to aircraft safety and performance (Hancz G., “CMMs in aerospace component manufacturing”, 2024) [2].

The dimensional inspection of these components generally relies on specific fixtures designed to replicate functional assembly conditions. The present study was conducted at LAUAK Group (FRANCE site), a major aerospace subcontractor supplying structural assemblies to OEMs such as Airbus, Safran, and Dassault.

Among the critical parts manufactured and inspected on this site is the sealing flange support, a machined ring-shaped component that connects the compressor stages of the LEAP-B1 engine, which powers the Boeing 737 MAX. This part must comply with strict flatness tolerances to ensure its functional integration within the engine.

In this study, the term sealing flange support refers to a circular metallic component subject to severe dimensional requirements, particularly in terms of flatness.

The inspection is carried out using a CMM supported by a self-supporting fixture composed of a ring resting on eight low-stiffness springs and held in place by twenty peripheral clamps, in order to replicate real assembly conditions. The entire assembly is mounted on a 300 mm rotary table.

However, repeated deviations from the specified flatness tolerance of 0.025 mm have been observed, occasionally reaching 0.15 mm. These discrepancies raise questions about the mechanical efficiency of the fixture and its ability to maintain geometric stability during inspection. Experimental studies have shown that clamping conditions can induce internal stresses (Von Mises) and significant deformations in machined parts, thereby compromising dimensional accuracy (Kolesnyk V. et al., “Experimental Evaluation of Flexible Fixture Stiffness for Steering Knuckles When Loading a Milling Machine Tool”, 2024) [3].

1.2 Problem Statement

The current inspection setup does not ensure reliable or repeatable flatness measurements. Although the elastic configuration of the fixture was designed to replicate a free resting condition, it induces deformations under clamping loads, thereby compromising measurement quality. These deformations lead to frequent non-conformities and necessitate systematic concessions from the client, with significant economic and schedule-related consequences.

In this study, the term self-supporting ring refers to a 600 mm diameter fixture placed on a 300 mm rotary table. The "suspended" sections describe its self-supporting behaviour.

Recent research has highlighted the limitations of overly flexible fixtures, whose mechanical response can become unpredictable and introduce non-negligible measurement bias. One study in particular shows that fixture flexibility, combined with clamping forces and friction, can compromise positioning stability, leading to significant systematic variations (Duret D. et al., "An investigation of indicators for controlling the quality of a fixture", 2010) [4].

Furthermore, a direct competitor has developed a reinforced fixture capable of meeting flatness specifications without requiring client concessions, thereby increasing pressure on the site to propose a robust alternative.

Research Question:

To what extent can the new fixture ensure the reliable and repeatable dimensional conformity of sealing flange supports?

1.3 Objectives and Scope of the Study

The primary objective of this study is to design and validate a metrology fixture capable of limiting deformation to below 0.025 mm, while complying with the technical and ergonomic constraints of the industrial context.

The specific objectives are as follows:

- Analyse the mechanical limitations of the existing fixture using finite element models
- Design several reinforced alternatives (ribs, internal rings, local overthicknesses)
- Simulate and compare the performance of each geometry
- Optimise both mass and stiffness to ensure conformity without compromising ergonomics

The study is limited to the mechanical behaviour of the ring under quasi-static loading. It does not include optimisation of the measurement program nor analysis of the CMM data. The deliverables include validated numerical models, simulation reports, and design recommendations.

1.4 Thesis Structure

This thesis follows the IMRaD structure, which is widely adopted in scientific research work:

- Chapter 2 presents a targeted literature review on the different domains relevant to this project, including dimensional metrology, fixture design, and mechanical simulation
- Chapter 3 describes the methodology, including modelling choices and the numerical analysis protocol
- Chapter 4 presents the simulation results, the comparison of variants, and performance validation
- Chapter 5 provides a critical discussion of the results, their limitations, and potential future work
- Chapter 6 concludes the study by summarising the main contributions and suggesting possible industrial implementation paths

This first chapter has introduced the technical background, outlined the problem, defined the research objectives, and delineated the scope of the study. The next chapter reviews existing research in the technical and scientific fields related to the problem.

Beyond the immediate industrial case, the methodology adopted in this thesis, combining structured problem solving (DMAIC), parametric modelling, and mechanical simulation has the potential to be applied to other dimensional inspection challenges involving flexible or geometrically complex parts. By providing a replicable approach to fixture optimisation, this work contributes to improving measurement accuracy in high-precision manufacturing while reducing reliance on concessions. It also aligns with broader trends in digital validation and predictive quality management within the aeronautical sector.

2 Literature Review

2.1 Introduction

Dimensional accuracy is a critical requirement in advanced industrial manufacturing, ensuring that components conform to design specifications, remain interchangeable, and function reliably within assemblies. In this context, inspection fixtures play a pivotal role. Beyond simply supporting parts during measurement, they contribute to reducing uncertainty and ensuring consistency by replicating representative boundary conditions. Their importance is amplified in modern applications where components are becoming increasingly lightweight, flexible, and sensitive to deformation. Even minor elastic distortions during inspection can lead to apparent non-conformities, particularly in the case of self-supporting geometries.

This literature review explores recent advances in the design and validation of three-dimensional inspection fixtures. It covers both mechanical and digital approaches, including parametric CAD modelling, finite element simulation (FEM), and stiffness optimisation strategies. Particular emphasis is placed on the challenges associated with the inspection of self-supporting rings under ultra-low deformation thresholds, for which conventional fixturing strategies often prove inadequate. The review aims to identify gaps in existing methods and tools, thereby establishing a clear rationale for the research framework adopted in this study.

This review is structured in thematic sections reflecting key design stages, from standard frameworks to material integration. Each section synthesises key literature and concludes with identified gaps. A structured evaluation grid (Appendix 2) complements this review by enabling a cross-source comparison.

2.2 Industrial Dimensional Control

2.2.1 Standards and Precision Requirements

Dimensional control in industry is governed by strict standards, among which ISO 1101 and ISO 10360 are particularly significant. The standard ISO 1101, officially referenced as AFNOR NF EN ISO 1101 *"Spécification géométrique des produits (GPS) - Tolérancement géométrique"* [5], defines geometric tolerancing by specifying symbolic and numerical requirements directly on technical drawings. It covers a wide range of tolerances: flatness, parallelism, perpendicularity, coaxiality, each expressed through standardised symbols followed by allowable values, as illustrated in Figure 1 [6]. These specifications ensure interoperability between design, manufacturing, and inspection. However, despite its foundational role, ISO 1101 remains largely descriptive and does not address practical aspects such as fixturing methods or deformation control during part clamping.

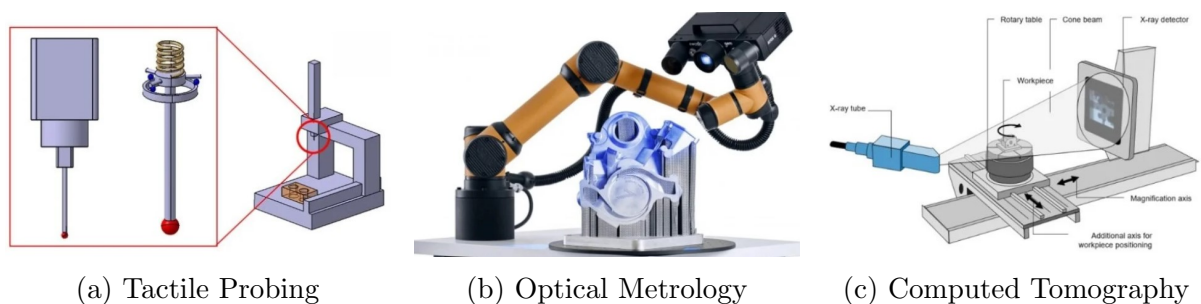
Complementary to design specifications, ISO 10360 focuses on performance verification of coordinate measuring machines (CMMs). This is notably addressed in the industrial guide *"ISO 10360 - The New Acceptance and Reverification Tests for Coordinate Measuring Machines"*, published by ZEISS [7]. The document highlights that dimensional accuracy depends not only on the measuring equipment but also on the applied measurement strategy and fixturing setup. As shown in Figure 2 [7], the guide demonstrates how

Measurement uncertainty reduced through many measured points

The graph shows a curve representing measurement uncertainty decreasing as the number of points increases. The y-axis is labeled 'Measurement uncertainty' and the x-axis is labeled 'Number of points'. A horizontal dashed line separates the 'Uncertain' region (above) from the 'Certain' region (below). A vertical dashed line at 10 points marks the 'Typical single-point probing range'. A vertical dashed line at 'Several hundred' points marks the 'Typical scanning range'. A vertical dashed line at 'Several thousand' points marks the 'Typical scanning range'.

Taken together, these two normative documents underscore the foundational importance of dimensional specification and verification in modern manufacturing. However, they also reveal a shared limitation: neither explicitly addresses the challenges of fixturing for deformation-sensitive parts. ISO 1101 provides the semantic and symbolic backbone for tolerancing, while ISO 10360, through ZEISS’s interpretation, formalises machine verification practices. Yet both documents overlook how mechanical compliance at the fixturing interface can compromise measurement integrity. This highlights a critical gap at the intersection of metrology, standards, and fixture engineering particularly relevant for applications requiring ultra-high precision.

Dimensional verification in manufacturing relies on multiple technologies, each adapted to specific part characteristics and inspection requirements. The most widespread techniques include tactile probing, optical metrology, and computed tomography (CT), illustrated in Figure 3 [8][9][10]. While tactile and optical methods dominate routine industrial control, CT remains restricted to niche applications due to its cost and complexity.



A detailed comparative perspective is provided by the article "*Differentiating Between Tactile and Optical Measurement Technology for Dimensional Inspection*", published by

Metrology.news in 2024 [9]. The article qualitatively contrasts tactile and optical technologies based on speed, precision, and surface compatibility. Tactile methods are presented as precise but invasive and time-consuming, while optical systems are positioned as faster and more flexible, but less effective for certain reflective or irregular surfaces. Despite its relevance, the article remains descriptive and lacks quantitative metrics or uncertainty analysis, limiting its applicability to critical inspection planning. Even so, it reinforces an essential idea: the inspection method must be chosen not only for its precision but for its interaction with the part, especially for deformation of sensitive components.

When compared to the structured measurement approach described in ZEISS’s ISO 10360 technical guide [7], which emphasises machine performance validation under controlled conditions, the Metrology.news article offers a broader technological overview but omits standardised evaluation criteria. Yet both converge on a fundamental insight: achieving accurate dimensional control is not solely a matter of instrumentation capability, it requires aligning the measurement strategy with surface characteristics, part stiffness, and fixturing constraints. This shared perspective underscores the need for integrated approaches that consider both metrological performance and mechanical sensitivity during inspection.

2.2.3 Gaps in Industrial Inspection

Based on the reviews of standards and measurement technologies, several methodological limitations can be identified.

Collectively, the standards and technologies reviewed above provide a solid foundation for dimensional inspection. ISO 1101 and ISO 10360 formalise geometric tolerancing and metrological validation, while tactile and optical systems offer flexible tools for geometry capture. However, these frameworks share a critical omission: they isolate measurement from its mechanical context.

Neither the standards nor the technologies adequately address how fixturing-induced deformation can compromise accuracy. Stable positioning is acknowledged, but guidance on managing compliance, clamping forces, or local stiffness is absent. As a result, inspection accuracy is seen as an outcome of instrumentation alone, neglecting mechanical interactions that can bias results.

This is particularly problematic for self-supporting components, where even minimal elastic distortion can breach tight tolerance limits.

2.3 Improving 3D Inspection Fixture Design

2.3.1 Positioning and Clamping Principles

Fixture design plays a key role in ensuring dimensional integrity, particularly for compliant components prone to elastic deformation. The CarrLane engineering guide *“Fixture Design Principles”* [11] outlines essential locating and clamping strategies, emphasising constraint-free positioning to minimise internal stresses. Poorly designed clamping is cited as a major source of deformation, especially under excessive forces or redundant

constraints. The widely adopted 3-2-1 principle, shown in Figure 4 [12], uses six contact points to fully constrain the part without over-restricting it. This method ensures both stability and adaptability to different geometries or access conditions. Yet, the CarrLane document does not address deformation thresholds $< 0,025$ mm, nor does it provide guidance for managing elastic responses under minimal clamping forces.

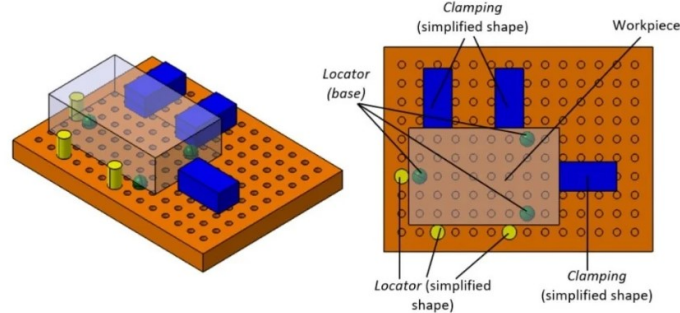


Figure 4: Ideal Fixturing System: 3-2-1 Principle

In contrast to these generic industrial guidelines, more advanced approaches tailored to deformation-sensitive applications have emerged in the scientific literature. A notable example is the research conducted by Shi et al., entitled *"Design and Manufacture of a Flexible Adaptive Fixture for Precision Grinding of Thin-Walled Bearing Rings"* [13]. The authors propose an adaptive fixturing system incorporating compliant supports and flexible elements, specifically designed to reduce elastic distortion in thin-walled parts during grinding and inspection. Their design is supported by finite element simulations and validated experimentally, demonstrating improved clamping performance and dimensional repeatability. While the methodology is robust, it remains focused on circular geometries and grinding operations, limiting its immediate transferability to broader industrial inspection contexts.

As a whole, these two sources illustrate the spectrum of fixture design strategies available, ranging from universal geometric principles such as CarrLane's 3-2-1 framework to tailored solutions like the adaptive system developed by Shi et al. The former offers simplicity and robustness but lacks specificity for ultra-low deformation applications, while the latter provides deeper insight into elastic behaviour management but with a narrower scope. This comparison reinforces the need for inspection tooling that combines mechanical reliability with local compliance, particularly when dealing with parts exhibiting high sensitivity to clamping-induced stresses.

2.3.2 Spring Systems and Adjustable Supports

Spring-loaded supports offer a compliant alternative to rigid clamping, particularly for geometrically variable or stress-sensitive components. They compensate for manufacturing tolerances and ensure soft contact without over-constraining the part. As illustrated in Figure 5 (left) [14], the spring compresses slightly to adapt to the real surface geometry, limiting internal stress while maintaining positional support. These functional principles are detailed in the technical article *"Precision Mechanical Springs for Critical Applications"*, published by St. Marys Spring Company [15]. The source outlines how

spring performance is governed by coil geometry, wire thickness, and material selection, all of which influence the load–deflection response and fixture sensitivity. Although not peer-reviewed, the article provides relevant design guidelines but does not address fixture integration in metrological contexts, nor the specific needs of self-supporting components where even minor reaction forces may distort the geometry.

Complementary structural insight is provided by Lin S. et al. in their research article *”Structure Design Improvement and Stiffness Reinforcement of a Machine Tool through Topology Optimization Based on Machining Characteristics”* [16]. The authors present adjustable locating devices, including topology-optimised structures designed to minimise parasitic forces during part immobilisation. As shown in Figure 5 (right) [16], these compliant designs enhance rigidity while adapting to complex geometries. The approach is validated through modelling and case studies, though it does not account for long-term degradation factors such as material ageing, thermal expansion, or fatigue behaviour under repeated use.

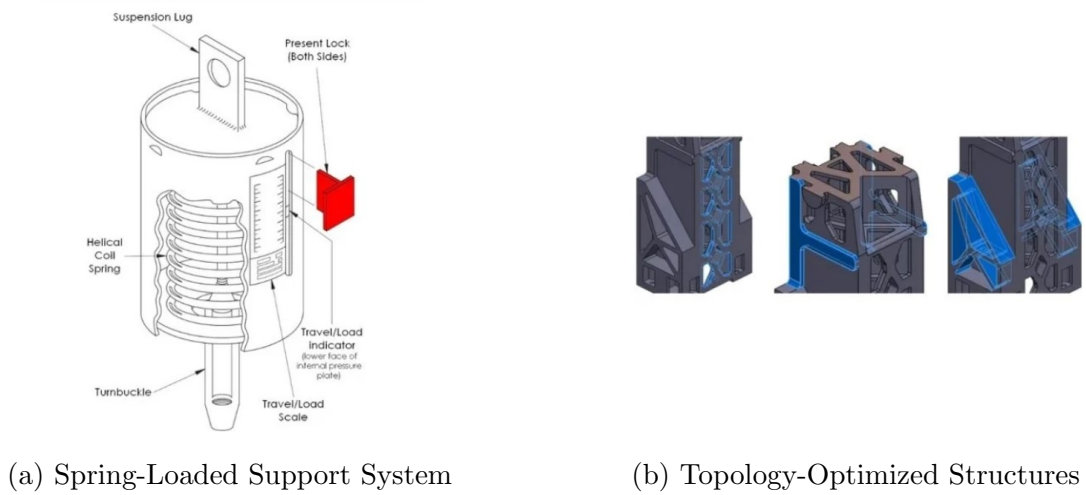


Figure 5: Adjustable Support Systems for Limiting Parasitic Forces

Taken together, these two sources demonstrate how compliance, whether introduced through discrete spring elements or through topology-optimised geometries, plays a critical role in reducing stress transmission during inspection. By limiting overconstraint while preserving positional accuracy, such systems help mitigate the risk of shape distortion. This is particularly relevant for self-supporting components, where the inspection setup must balance fixture adaptability with repeatable alignment. The present study builds on this rationale to explore configurations that combine mechanical decoupling with dimensional stability.

2.3.3 Gaps in Design Improvements

Based on the preceding review of fixturing principles and adjustable support systems, several key limitations emerge regarding current design approaches for inspection tooling.

Sections above outlined the range of strategies available from conventional rigid positioning systems to more compliant and adaptive solutions. The 3-2-1 fixturing principle

remains a universal and reliable baseline, while advanced adaptive designs and compliant mechanisms provide promising alternatives for minimising stress transfer. Nonetheless, these approaches tend to be considered independently, without a unified framework that integrates mechanical compliance, geometric precision, and fixture adaptability.

In particular, traditional design guides such as CarrLane lack quantitative guidelines for managing deformation-sensitive parts, while more sophisticated solutions like spring systems or topology-optimised supports offer local adaptability but are rarely validated in metrological contexts. Furthermore, existing studies often focus on part machining or handling, rather than the unique constraints imposed by dimensional inspection where repeatability, non-invasiveness, and alignment consistency are critical.

This gap is particularly relevant for self-supporting components, where the fixture must ensure accurate support without inducing elastic deformation. The present work responds to this challenge by exploring hybrid designs that integrate compliant elements within a rigid architecture, aiming to achieve both stress decoupling and geometric fidelity during inspection.

2.4 Parametric 3D Design with CATIA

2.4.1 The Role of CAD in Fixture Design

Computer-Aided Design (CAD) tools are central to the development of dimensional inspection fixtures, supporting accurate modeling, assembly, and documentation. CATIA V5 is particularly suited for this purpose thanks to its integrated environment for 3D part design and drafting. As outlined in the university material *"CATIA V5 Lectures"* by Dr. Kovacevic A. [17], the platform enables the creation of complete fixture assemblies, with manufacturing drawings and DXF exports handled within the same workflow. This integration promotes consistency across design and production teams, as illustrated in Figure 6 [18]. However, the document remains focused on general educational usage and does not address specific constraints related to deformation-sensitive parts like self-supporting component or high-precision inspection, limiting its direct applicability to metrology tooling.

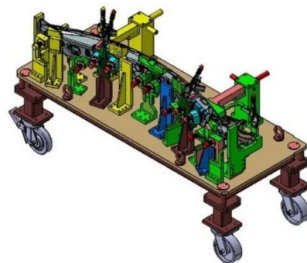


Figure 6: Complex Fixture Design - CATIA

2.4.2 Parametric Modeling and Associativity

Beyond general CAD usage, parametric modeling improves design adaptability, especially when inspection tooling must accommodate evolving part geometries. In their

article *"Parametric Design and Associativity in 3D CAD"* [19], Wang et al. emphasise how associativity and constraint-based modeling enhance consistency and efficiency across assemblies. In contrast, Kovacevic's practical lectures neglect compliance aspects, illustrating a persistent disconnection between educational design logic and metrology requirements. As shown in Figure 7 [20], dimension-driven parameters allow for automatic updates without redesign. Nevertheless, the publication remains focused on general CAD principles and does not consider specific challenges such as fixture compliance, surface adaptation, or deformation control, limiting its direct application to precision inspection systems.

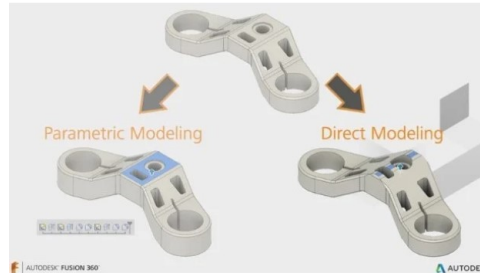


Figure 7: Parametric vs Direct Modeling

2.4.3 Gaps in CATIA Design

Collectively, these two sources highlight the importance of CAD tools in developing consistent and adaptable fixture models. CATIA V5 offers an integrated environment for design and documentation, while parametric modeling supports iterative updates during part and fixture evolution. Yet, both references remain general in scope and do not address critical aspects such as compliance management or deformation control. This limitation reinforces the relevance of a tailored, constraint-driven approach in the present study, combining parametric design with mechanical considerations specific to metrological tooling.

2.5 Digital Validation of 3D Fixtures Using FEM with ANSYS

2.5.1 General Principle of Finite Element Method

The finite element method (FEM) enables simulation of mechanical responses under load before physical manufacturing. In his technical manual *"Finite Element Simulation with ANSYS Workbench 2019"* [21], Lee H. presents FEM as a tool for predicting stresses, displacements, and flatness deviations in mechanical systems. As shown in Figure 8 [22], it can model how a beam deflects under applied forces. This capability is essential for detecting potential deformation risks in inspection fixtures early in the design process. However, while the book is comprehensive on simulation workflows, it does not address specific cases such as compliant or spring-loaded boundary conditions, which are highly relevant in dimensional metrology applications.

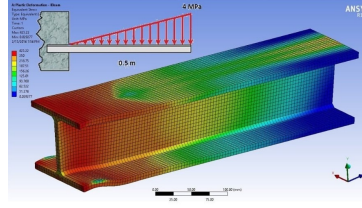


Figure 8: Beam Deformation Prediction - ANSYS

2.5.2 Boundary Conditions and Specific Cases

The predictive accuracy of FEM strongly depends on how boundary conditions are defined. As illustrated in Figure 9 [23], realistic constraints such as fixed supports, clamping forces, or flexible contacts must be carefully modeled. Lee [21] emphasises that boundary definition is critical for credible simulations, yet the guide assumes rigid or simplified conditions and omits complex support behaviors like compliant pads or spring elements. In inspection tooling, these omissions may lead to inaccurate estimations of part deformation, especially in self-supporting configurations. This limitation highlights the need to adapt standard simulation strategies when dealing with delicate components or adaptive fixtures.

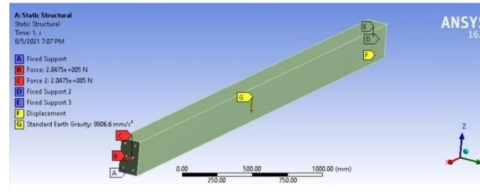


Figure 9: Beam Boundary Conditions - ANSYS

2.5.3 Mesh and Convergence Study

Mesh refinement plays a key role in FEM accuracy. Hemesh P. et al., in their article "*Mesh Convergence Study and Estimation of Discretization Error*" [24], demonstrate that stress and displacement predictions stabilise as mesh density increases. As shown in Figure 10 [25], convergence is evaluated by tracking variations across coarse, medium, and fine meshes. The study confirms that finer meshes reduce discretisation error but significantly increase computation time. Although highly relevant for validating numerical stability, the study does not explore constraints specific to inspection fixture design, such as modeling of contact stiffness or local compliance zones.

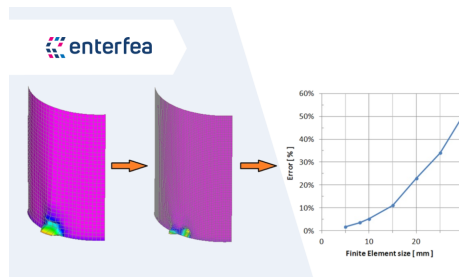


Figure 10: Mesh & Convergence - ANSYS

2.5.4 Gaps in ANSYS Simulation

These three perspectives collectively illustrate that robust digital validation depends not only on the availability of simulation tools like ANSYS, but also on two critical enablers: appropriate boundary conditions and mesh refinement. While Lee [21] provides a comprehensive overview of FEM use in mechanical validation, his methodology assumes idealised contacts and lacks consideration for metrology-specific constraints. Similarly, Hemesh et al. [24] detail the importance of mesh convergence but omit the implications of soft or adaptive supports. While the present study attempted to implement compliant boundary conditions using soft-contact models and spring elements, these configurations proved difficult to stabilise within the available ANSYS framework. As a whole, these sources highlight that accurate fixture simulation requires an integrated strategy accounting for both physical realism and numerical robustness, precisely the approach adopted in this work for the validation of compliant and self-supporting inspection fixtures.

2.6 Mechanical Behaviour of Fixture Materials

2.6.1 Properties of Tool Steels

The mechanical performance of inspection fixtures depends largely on the properties of the materials used in their construction. As illustrated in Figure 11 [26], characteristics such as Young’s modulus, yield strength, ductility, and machinability govern how materials respond to applied forces and how they maintain dimensional integrity under load. In their technical book *“Tooling Steel”* [27], Roberts G. et al. describe how tooling steels like P20 (40CMD8) are selected for their stiffness and resistance to deformation. High elastic modulus ensures structural rigidity, while adequate yield strength prevents plastic flow during clamping or loading. These attributes are essential to maintain fixture geometry during metrological use. Nevertheless, the reference focuses on general mechanical behaviour and lacks specificity regarding steels used in metrology-related applications, such as for deformation-sensitive or lightweight components.



Figure 11: Mechanical Properties - Materials

2.6.2 The Case of 40CMD8 (P20)

Among tooling steels, P20 (also known as 40CMD8) stands out for its balanced combination of strength, dimensional stability, and machinability. According to the technical ar-

ticle published by Mantle *"Tool Steels and Injection Modeling: A Comprehensive Guide"* [28], P20 offers excellent workability without requiring post-machining heat treatment. It is particularly suited for high-precision applications where repeatability and moderate mechanical strength are critical, as in the case of prototype tooling or compliant fixture designs. As shown in Figure 12 [29], its mechanical and chemical properties support its use in precision contexts. While the article focuses on injection moulding, the principles are transferable to dimensional inspection tooling, especially when adaptability and geometric fidelity are priorities. Nonetheless, the source remains limited in scope, offering little insight into the metrological implications of material behaviour during repeated loading or over time.

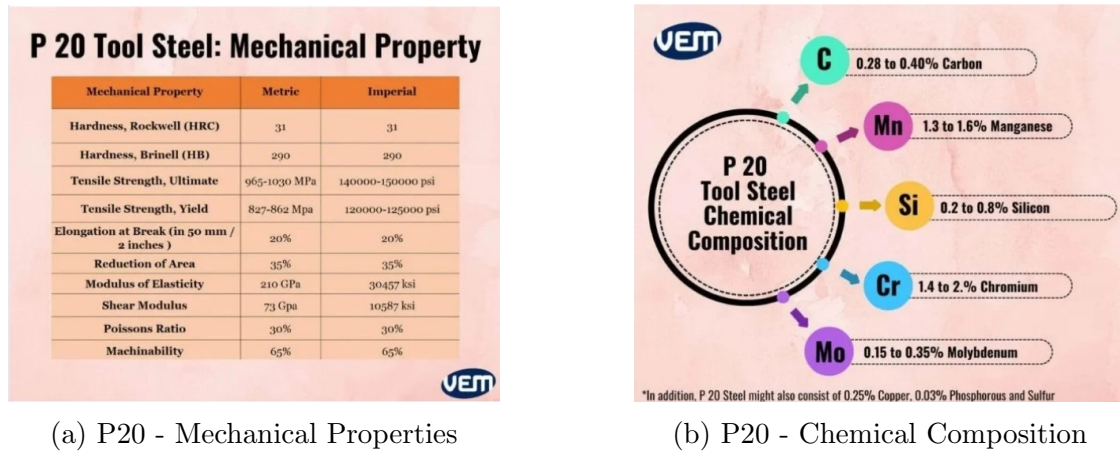


Figure 12: P20 / 40CMD8 Characteristics

2.6.3 Residual Stresses in Machining

Machining is a widely used process in the manufacturing of control fixtures due to its ability to shape precise geometries from metallic blocks. However, it introduces complex internal effects within materials, notably residual stresses. As illustrated in Figure 13 [30], thermal and mechanical phenomena generated during cutting, such as tool pressure, chip formation, and localized heating, produce residual stress fields within the part. These stresses, if not controlled, may lead to post-machining distortions that compromise dimensional conformity and repeatability.

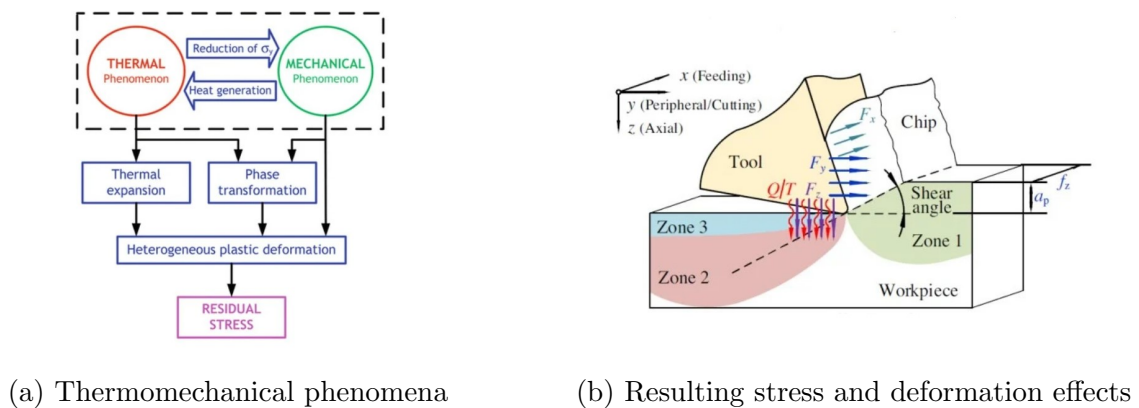


Figure 13: Mechanisms of Residual Stress Generation during Machining

To better understand and mitigate these effects, Weber et al. conducted an experimental study focusing on milled aluminium workpieces [31]. Using X-ray diffraction, they demonstrated that residual stresses depend heavily on process parameters, such as tool path strategies and cutting speeds. Their findings confirmed the lack of repeatability in stress distribution and resulting part distortion, highlighting a critical challenge for ensuring dimensional integrity in machined components. Nevertheless, their research remains limited to aluminium alloys and cannot be directly transposed to steels like 40CMD8, commonly used in fixture design.

In contrast, Soori et al. provided a comprehensive literature review on machining-induced residual stress [30]. Their analysis covered a broad spectrum of influencing factors, including material properties, tool geometry, cutting strategies, and cooling methods. The authors emphasized that these parameters interact to define the final stress profile within the workpiece, and suggested potential mitigation strategies. Although the review lacks original experimental data and offers limited quantitative comparisons, it proposes a valuable synthetic overview of the state of the art, relevant for steel-based applications.

Overall, both studies underscore the importance of understanding residual stress mechanisms in order to improve the accuracy of machined control tools. While Weber et al. provide experimental validation for specific cases, Soori et al. offer a theoretical framework that may inform the selection of machining strategies better suited to fixture materials like P20 (40CMD8). A gap remains, however, regarding targeted studies on residual stresses specifically in dimensional control fixtures using tool steels under realistic industrial conditions.

2.6.4 Gaps in Tool Steel Behaviour

Taken together, these four sources highlight both the mechanical and process-related aspects critical to the use of tool steels in inspection fixtures. Roberts et al. [27] provide a metallurgical foundation for understanding the stiffness and yield strength requirements of fixture materials, while Mantle [28] adds a practical perspective on their machinability and precision performance, particularly for P20 (40CMD8). Weber et al. [31] experimentally demonstrate how residual stresses introduced during milling operations can compromise dimensional accuracy, whereas Soori et al. [30] offer a comprehensive synthesis of the factors influencing such stresses and suggest mitigation strategies applicable to steel alloys.

Yet, several knowledge gaps remain. None of these sources directly investigate the behaviour of 40CMD8 under the specific conditions of metrological use—namely, under repeated low-stress load cycles typical of inspection operations. Furthermore, they do not assess how such materials perform when integrated with compliant or spring-loaded support systems, which are often used in adaptive fixturing to minimize induced deformation. There is also a lack of experimental data validating the residual stress profiles and resulting deformations in actual steel-based dimensional control tools.

This reinforces the need, in the present study, to go beyond static property evaluation. Material choices must be validated not only in terms of strength and machinability but

also in relation to their dynamic response, residual stress sensitivity, and suitability for integration within compliant fixture designs used in aerospace metrology.

Beyond material selection and machining-induced behaviour, fixture performance also depends on structural and mechanical strategies to control stiffness and adaptability.

2.7 Optimizing Fixture Stiffness

2.7.1 Structural Techniques

Improving fixture stiffness is essential to minimise deformation during dimensional inspection. Common strategies include adding ribs, reinforcement zones, or applying topology optimisation to remove non-functional material while maintaining structural integrity, as illustrated in Figure 14 [32][33]. Lin S. et al. [16] demonstrate that such enhancements increase stiffness without adding weight, which is key for ergonomic handling. However, these structural features are fixed once designed, limiting adaptability to part variability or clamping-induced distortion. For self-supporting or deformation-sensitive components, this rigidity can cause overconstraint or misalignment, particularly in prototyping. This underlines the need for more flexible stiffness strategies that balance rigidity with controlled compliance.

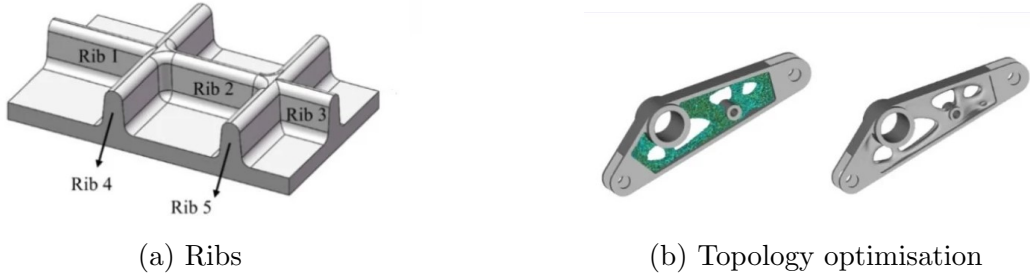
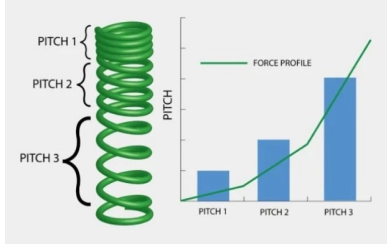


Figure 14: Methods to Improve Fixture Stiffness

2.7.2 Mechanical Adjustability

To address the limitations of fixed reinforcement, mechanical adjustability offers a flexible alternative for managing fixture stiffness. Systems such as variable-stiffness springs, calibrated shims, and micrometric screws illustrated in Figure 15 [34][35][36] enable fixtures to adapt to minor geometric variations, especially in prototyping or low-volume contexts. Chase W. et al. propose in their article *"Design of a Variable Stiffness Spring with Human-Selectable Stiffness"* [37] a manually tunable spring using a selector mechanism and discrete anchoring points, allowing stiffness adjustment without component changes. Originally developed for biomechanics, this concept is transferable to metrology, offering adaptable contact forces and reducing deformation risks. Nevertheless, such systems may require adaptation to meet the mechanical stability and repeatability constraints of dimensional inspection setups.



(a) Variable-stiffness spring



(b) Shims



(c) Micrometric screws

Figure 15: Adjustable Stiffness Systems

2.7.3 Gaps in Stiffness Optimisation

The structural and mechanical strategies presented above offer complementary benefits: structural reinforcements improve stiffness with low weight, while adjustable systems enhance adaptability to part variability. Yet, each has limitations when used alone. Rigid structures may cause overconstraint in sensitive parts, while adjustable mechanisms can lack the robustness needed for high-precision inspection. This highlights the need for hybrid solutions combining structural integrity with controlled compliance. The present study builds on this approach to design fixtures suited for accurate and repeatable inspection of self-supporting components.

2.8 Identified Gaps in Literature

The literature reviewed across standards, design methodologies, digital validation tools, material selection, and stiffness optimisation has provided a robust foundation for this study. Each source contributed to broadening the understanding of fixture design and dimensional control. Nonetheless, a transversal analysis reveals that most references, while technically relevant, fall short when applied to the specific context of self-supporting components requiring ultra-low deformation (<0.025 mm).

Several critical limitations emerge:

In industrial inspection, existing standards and measurement strategies consider accuracy primarily as a function of instrumentation. They rarely account for how fixture-induced deformation can compromise results, particularly in compliant or deformation-sensitive parts.

In fixture design improvements, conventional systems (e.g., 3-2-1 layouts) ensure geometric control but offer no adaptability to local compliance. Adaptive or spring-loaded supports exist but lack validation in high-precision inspection scenarios, often remaining in conceptual or biomechanical domains.

In CAD modeling, while platforms like CATIA and parametric design principles enable geometric flexibility, they do not inherently integrate compliance logic or deformation control, limiting their standalone relevance for metrology tooling.

In digital validation with FEM (ANSYS), general simulation workflows are well established, yet fail to model soft supports, clamping flexibility, or the boundary conditions

unique to metrological fixtures, factors crucial for accurate predictions under minimal load.

In material selection, steel grades like 40CMD8 are mechanically suitable, but literature does not address their behaviour under repeated low-stress cycles or their integration with compliant features. Moreover, the impact of machining-induced residual stresses on dimensional stability is often overlooked, despite its relevance for precision fixtures.

In stiffness optimisation, structural reinforcement ensures rigidity but lacks adaptability, while adjustable systems introduce flexibility at the cost of mechanical robustness. Neither fully resolves the conflicting needs of precision, repeatability, and compliance.

Viewed together, no existing study proposes a unified methodology that combines parametric CAD, metrology-aware FEM simulation, and hybrid stiffness strategies tailored to the inspection of self-supporting components under ultra-low deformation. This multidimensional gap justifies the approach adopted in this research, which aims to develop an innovative, constraint-driven fixture design framework suited for high-precision, low-distortion dimensional control.

2.9 Source Review Grid

In accordance with Method 1 outlined in the writing guidelines, each source analysed in this review has been synthesised using a structured evaluation grid. This table, available in the Appendice 2, compiles key information for each reference, including the main arguments, methodological approaches, conclusions, thematic relevance, and a critical appraisal of identified limitations. It provides a coherent comparative framework for assessing the relevance and contribution of each source to the present research.

2.10 Conclusion

This literature review has examined the key domains underpinning dimensional inspection in industrial contexts, including geometric standards, measurement technologies, fixture design strategies, CAD-based parametric modelling, finite element simulation, and material selection. Through a critical analysis of over a dozen academic and industrial sources, it revealed a consistent limitation across these fields: the absence of a unified methodological framework specifically tailored to the inspection of self-supporting components under ultra-low deformation constraints.

While standards such as ISO1101 and ISO10360 establish the metrological foundations, they do not account for deformation risks linked to fixturing. Similarly, current design practices often dissociate structural stiffness from mechanical adaptability, neglecting the need for compliant yet stable support systems. Digital tools such as CATIA and ANSYS offer powerful capabilities, but are rarely applied in an integrated way to simulate and validate the behaviour of inspection fixtures under realistic conditions in particular soft-springs. Finally, although suitable materials such as 40CMD8 exhibit strong mechanical properties, their behaviour within adaptive or spring-loaded systems remains insufficiently explored.

Taken together, these gaps fully justify the rationale behind the present study. By combining parametric CAD design, advanced FEM simulation, and the development of hybrid fixtures integrating structural and compliant features, this research aims to address the unresolved challenges of dimensional verification for deformation-sensitive, self-supporting geometries. The following sections will detail the methodology, experimental setup, and design process adopted to implement and validate this approach.

The following chapter presents the methodology developed to address the identified gaps and to define the most appropriate strategy for this iterative engineering project.

3 Methods

This section outlines the methodological approach adopted to design, model, and validate a dimensional inspection fixture intended to ensure the geometric conformity of a critical aeronautical component. The objective is to ensure scientific rigour, reproducibility, and traceability of the process, by justifying the choices made at each step. The methodology is based on the DMAIC cycle, commonly used in industrial problem-solving approaches, while also adhering to the scientific standards of the IMRaD structure required for this study.

To anchor this process in a structured research framework, the “*Research Onion*” model (Saunders et al., 2019) [38] was used, particularly its dimensions related to methodological choices, data collection, and numerical analysis.

3.1 Methodological Framework

As this project addresses a concrete industrial problem, a DMAIC-type approach was chosen to structure the process. This methodology is particularly suited to iterative design contexts, as it allows for continuous improvement based on successive validations.

The process began with an assessment of the current fixture, based on field observations and real-world data. This helped to clarify the actual needs and led to the formulation of the problem statement. Then, a range of potential solutions was explored, aiming to fully or partially address the identified issues. Finally, each solution was evaluated against the initial set of technical and performance requirements, in order to determine whether it met the expected level of reliability and conformity.

This iterative structure was complemented by a question-driven logic, used at each stage to guide the design choices and validation steps in a systematic and critical manner. Accordingly, the Results and Discussion are presented in a single integrated chapter, as this structure better reflects the continuous reasoning process and enhances the clarity and coherence of the narrative.

3.2 Definition of Technical Requirements

The technical requirements were defined based on the functional analysis of the fixture and the constraints of the inspection process. The fixture had to meet the following criteria:

- **Mass:** The additional weight was not to exceed 18 kg compared to the current fixture, in order to remain within the rotational table’s capacity. The ergonomic limits are defined by AFNOR standard NF X35-109 [5] (assisted handling beyond 25 kg).
- **Geometry:** Access to all 160 probing points had to be maintained without modifying the existing CMM program or requiring new probe acquisition. The final flatness was expected to remain below 0.025 mm, with a clearance height of 39.57 mm $\pm$ 0.1 mm.

- **Robustness:** The structure had to be compatible with 40CMD8 tool steel, incorporating internal geometric reinforcements to improve stiffness without excessive weight.
- **Environment:** The fixture had to remain compatible with its inspection environment (rotating table and eight soft springs), while ensuring measurement stability and repeatability.

These criteria were translated into two key performance indicators: the mass of the self-supporting ring and its maximum deformation under quasi-static load.

3.3 Modelling Assumptions

Two experimental configurations observed on site were used to define the simulation hypotheses:

- **Eight-spring configuration:** : This represented the standard operating condition, where the ring rests freely on soft supports.
- **Three-support configuration:** : Used as a worst-case scenario, this setup reflected the most severe flatness deviations observed in real conditions. It was chosen to test the fixture’s performance under degraded support conditions.

The simulation was based on a linear, elastic, and isotropic material behaviour in a static regime, ignoring dynamic and plastic effects. This choice was motivated by the need to keep the model simple while maintaining sufficient representativeness for comparing design variants. As noted by Lee H. in his studies on finite element simulation in ANSYS [21], simulation accuracy primarily depends on boundary conditions and modelling coherence.

3.4 Meshing and Convergence Study

The numerical models were developed in ANSYS Workbench using CAD geometries imported from CATIA V5. A convergence study was conducted to determine the optimal mesh size that would ensure a suitable balance between accuracy and computational cost. This approach is supported by the work of Hemesh P. et al. [24], who emphasised the importance of progressive mesh refinement to stabilise displacement and stress predictions.

3.5 Variant Developments

Based on the technical requirements and prior hypotheses, several geometric variants were modelled to test different stiffness optimisation strategies:

- **Ribs** to increase moment of inertia without adding significant mass.
- **Extra Thickness** to increase rigidity without modifying the part’s geometry.
- **Internal Reinforcements** to stabilise the ring without altering thickness

These strategies, analysed by Lin S. in the context of structural optimisation using geometric reinforcements [16], reinforce the relevance of the approach adopted in this study.

These variations, involving changes in thickness, width, and height, are particularly well-suited to parametric modelling in CATIA. As stated by Wang et al. in their article [19], this approach allows for geometrical consistency while facilitating iterative design adjustments. This justifies the use of this design environment for the purposes of this study.

In parallel, several tooling materials were considered as alternatives to 40CMD8, based on their Young's modulus, density, and machinability. As highlighted by Robert G. in his reference work on material selection [27], mechanical properties (especially stiffness) remain the primary performance criteria for structural industrial components.

3.6 Approach Validation

Validation relied on two main pillars:

- A satisfactory correlation between simulation predictions and real-world flatness measurements from the current fixture
- Formal approval of the technical approach by the industrial client (SAFRAN), including validation of the protocol and performance criteria

This dual validation, both numerical and industrial, confirms the relevance of the modelling approach adopted.

3.7 Limitations and Simplifications

The simplifications applied in this model (linear behaviour, absence of friction, idealised boundary conditions) were part of a comparative logic aimed at differentiating design variants. However, they also reflect an intention to explore solutions that are rarely addressed in the current literature while remaining within a realistic and reproducible industrial framework.

As demonstrated in the cross-sectional analysis of the literature review, very few studies combine parametric modelling in CATIA, ANSYS simulations adapted to low-deformation self-supporting components, and comparative analysis of geometric reinforcements designed for metrological applications. The present approach attempts to partially bridge this gap by evaluating concrete variants (ribs, internal reinforcements, thickened sections) in a mass/stiffness optimisation context, without resorting to complex, non-industrialisable models.

Although the model does not account for nonlinear effects or contact phenomena, it serves as a pragmatic contribution to the development of dimensional control solutions that are reliable, repeatable, and compatible with the operational constraints of industrial inspection.

3.8 Reproducibility

All CAD files, simulation configurations, material properties, and modelling assumptions were documented and archived. All data are stored on the industrial site’s collaborative workspace, ensuring traceability, reusability, and reproducibility within a broader industrial context.

3.9 Ethical Considerations

No part of this study poses a risk to the physical or moral integrity of individuals. The project was classified as Category 1 risk under the CURES process, and the approval letter is provided in Appendix 1. All data used (drawings, materials, tolerances) originate from non-confidential internal documents and were handled in accordance with academic and industrial ethical standards.

3.10 Conclusion

The methodology presented combines scientific rigour, industrial relevance, and reproducibility. It relies on a DMAIC structure mapped onto the IMRaD format, with systematic justification of technical choices and dual validation, both numerical and industrial.

Beyond its comparative purpose, this approach addresses concrete industrial challenges related to dimensional conformity, by exploring innovative geometric variants (ribs, internal rings, thickened regions) through parametric modelling in CATIA and numerical analysis in ANSYS. While not intended to push the boundaries of advanced modelling (nonlinearities, contact phenomena), it offers an original methodological contribution by combining implementation simplicity, industrial reproducibility, and the ability to test solutions that are rarely documented in academic literature.

It thus provides a structured foundation for the exploration of potential solutions and the critical analysis of results presented in the next chapter, with the aim of addressing the issue of dimensional conformity in self-supporting components subjected to extremely tight flatness tolerances.

4 Results & Discussion

4.1 Introduction

This chapter presents the results of the proposed dimensional inspection fixture design, together with a critical analysis structured according to the DMAIC-based methodology introduced earlier. Each result is interpreted in terms of its technical significance and evaluated against the objectives defined in the Methods section. The reasoning follows a question-driven approach, focusing on the key challenges identified throughout the project.

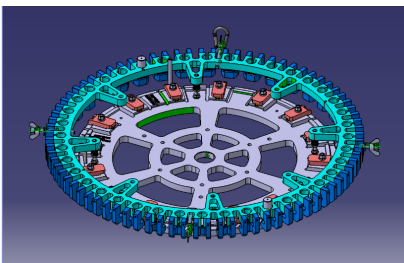
In this context, the Results and Discussion are presented jointly to reflect the continuous flow between design iterations, technical assessments, and engineering decisions. This structure facilitates a clearer understanding of how each solution was explored, developed, and validated, based on empirical observations, numerical simulations, and relevant benchmarks.

4.2 Define

Before investigating potential design improvements, it is essential to clearly define the functional requirements, the operating principles of the existing inspection fixture, and its current performance relative to the specified objectives. Particular attention is given to identifying the sources of deviation observed during actual use. This preliminary step provides the necessary engineering overview to ensure that any optimisation effort remains grounded in operational reality and fully aligned with the needs of the inspection process.

4.2.1 Current Inspection Fixture and Operation

As introduced in the first chapter, LAUAK currently encounters difficulties with the dimensional control of a sealing flange support component, which connects two sections of the LEAP-B1 engine used on the Boeing 737 MAX. Both the fixture and the sealing flange are illustrated in Figure 16. This component provides both mechanical flexibility and structural integrity to the assembly, while supporting the blades of a high-pressure compressor stage.



(a) Fixture



(b) Sealing flange support

Figure 16: Inspection Fixture and Sealing Flange Support

Dimensional inspection is performed via tactile probing on 160 control points distributed across 40 fins (4 points per fin), as shown in Figure 17.

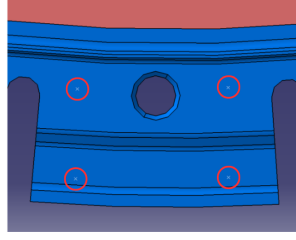
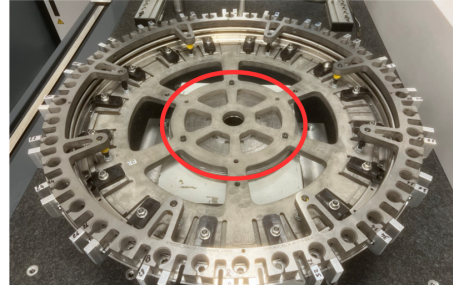


Figure 17: Probing Points

The fixture is mounted on a $\varnothing 300$ mm rotary table. The sealing flange rests solely on its internal diameter, highlighted in red in Figure 18, which necessitates the use of a self-supporting crown.



(a) CMM with rotary table



(b) Fixture in position

Figure 18: Installation of the Fixture on the CMM

The crown is simply laid on eight low-stiffness compression springs that provide no real constraint force, as shown in Figure 19. The springs rest on the counterbored holes of the crown.

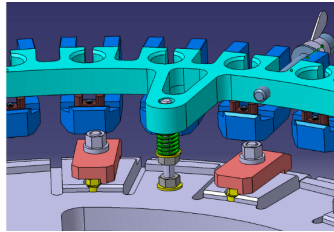


Figure 19: Low-stiffness Springs (x8)

The flange is clamped to the crown using twenty peripheral brackets (instead of the initially planned forty), with a tightening torque of 3 Nm, as illustrated in Figure 20.

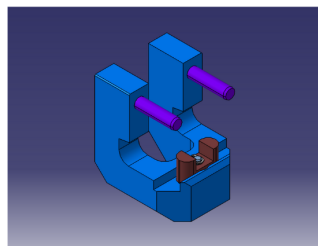


Figure 20: Clamping Bracket

To provide a synthetic overview of the current fixture’s operating principles and highlight the key functional relationships, a functional diagram is presented in Figure 21.

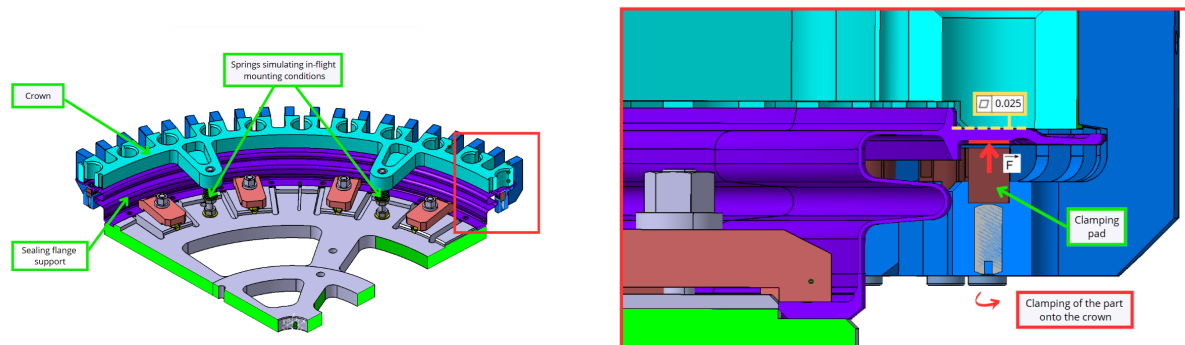


Figure 21: Functional Diagram of the Fixture

This schematic representation summarises the main elements previously described, including the support strategy, clamping method, and structural interactions involved in the inspection process.

4.2.2 Observed Issue

Once it was confirmed that the fixture did not enable reliable dimensional control of the part, a dedicated test campaign was conducted. The objective was to identify the potential sources of error by analysing the behaviour of the self-supporting crown alone, under different support configurations. This approach aimed to determine whether the deformation was primarily due to the crown’s geometry and its insufficient stiffness or to an external reason.

The flatness measurements obtained from this campaign are summarised in Table 1.

Table 1: Flatness of the Self-supporting Crown - Various Support Conditions

Configuration	Flatness
Crown placed on the control surface plate (granite table)	0.029 mm
Crown placed on 8 springs (within the fixture)	0.162 mm
Crown placed on 4 springs (within the fixture)	0.220 mm
Crown placed on 3 rigid supports (pillars)	0.250 mm

The inspection results revealed significant flatness deviations depending on the support configuration. The measured flatness deteriorated as the number of contact points decreased, ranging from 0.162 mm with 8 springs to 0.250 mm when supported on only three rigid points positioned at 120° intervals.

For reference, the required flatness tolerance is 0.025 mm when the crown is mounted on the 8-spring configuration, meaning the observed deviation exceeds the acceptable

limit by approximately 0.125 mm.

These results confirm that the current fixture cannot ensure dimensional compliance under consistent operating conditions.

4.2.3 Root Causes Identified

The analysis highlighted two main root causes. First, the flatness defect observed in the configuration with three rigid supports confirms that the crown deforms under its own weight, which is indicative of insufficient structural stiffness.

Second, when supported by four or eight springs, the crown exhibited irregular contact and unstable planarity. This suggests that the springs may not have been of uniform height, introducing local variations in support and contact force. Although the spring heights were verified using a height gauge, the measurement precision may not have been sufficient given the extremely tight customer flatness tolerance.

These deviations are therefore partly intrinsic to the crown's structure, and partly caused by external setup conditions. In order to confirm this hypothesis, a full inspection of the 160 probing points has been done using the 8-spring configuration. The height coordinates of each measured point were then mapped and imported into ANSYS for surface reconstruction, as shown in Figure 22.

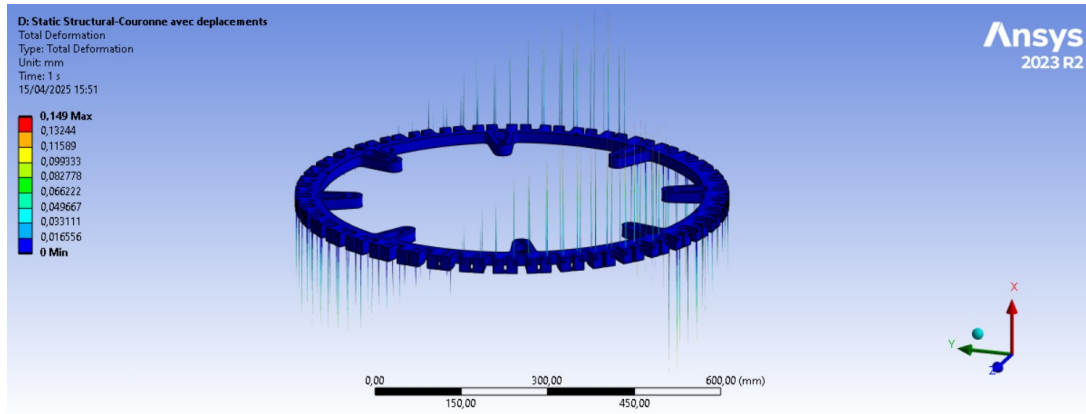


Figure 22: Displacement of the Self-supporting Crown - ANSYS

The shape of the crown is clearly visible, with peak deviations reaching ± 0.15 mm relative to the nominal height of 39.57 mm.

To go further, a targeted analysis was performed by isolating the average height values (based on the four probing points per fin) at the locations corresponding to the eight springs. This made it possible to quantify the local deviation at each spring contact zone. The measurements are provided in Table 2.

Table 2: Average Height of Springs

Spring No.	Average Z Coordinate	Deviation from Nominal (39.57 mm)
1	39.654 mm	+0.084 mm
2	39.5765 mm	+0.0065 mm
3	39.6303 mm	+0.0603 mm
4	39.635 mm	+0.065 mm
5	39.471 mm	-0.099 mm
6	39.441 mm	-0.129 mm
7	39.6086 mm	+0.0386 mm
8	39.697 mm	+0.127 mm

The results show clear differences between support heights, confirming that spring height variation contributes significantly to the overall flatness defect. The most significant deviations were measured at -0.129 mm and $+0.127$ mm, relative to the nominal height. The hypothesis of non-uniform spring heights is therefore validated.

4.2.4 Summary of Findings

In line with the observations presented in the Introduction chapter, it was confirmed that the current fixture is not capable of ensuring the dimensional validation of the sealing flange support, leading to significant industrial implications such as production blockage or conditional part acceptance through client waiver processes. These procedures are administratively time-consuming and not viable in the long term. The project therefore focused on improving the self-supporting crown, illustrated in Figure 23, identified as the most influential and critical subassembly within the overall fixture.

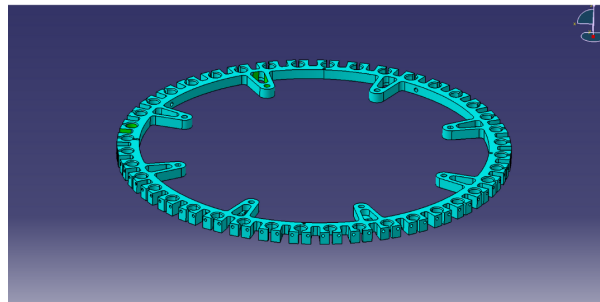


Figure 23: Current Self-supporting Crown - CATIA

In addition, part of this study aims to address the issue of spring height variation, which was found to significantly affect the flatness of the crown during inspection.

Rather than redesigning the entire fixture, the DMAIC approach was applied specifically to the crown, with the objective of reinforcing its mechanical stability while preserving both functional compatibility and manufacturing constraints.

4.3 Measure/Model

Once the fixture's operation was described, its deviations from the initial objectives identified, and the scope of the study refocused on the self-supporting crown, it became possible to explore potential improvement strategies. These are guided by intermediate research questions derived from the performance indicators defined in the Methods section.

However, before any design iteration, it is essential to develop a numerical model that accurately reproduces the real mechanical behaviour of the crown. For this purpose, several hypotheses were formulated, and a mesh convergence study was conducted to validate the reliability of the ANSYS simulation model.

4.3.1 Hypotheses

Two configurations were modelled:

- A standard case, representing the typical use of the crown resting on eight springs.
- An extreme case, simulating an unfavourable scenario where the crown rests on only three rigid supports positioned at 120° from each other.

Both configurations share the same fundamental assumptions to ensure consistency between simulations:

- **Material:** 40CMD8 steel, with the following mechanical properties:
 - Young's modulus: 205 000 MPa
 - Density: 7 850 kg/m³
 - Poisson's ratio: 0.3
- **Applied force:** a vertical load of 111.5 N, corresponding to the weight of the crown ($11.37 \text{ kg} \times 9.81 \text{ m/s}^2$)

The main difference between the two configurations lies in the boundary conditions.

In the eight-spring setup, the springs are considered undeformable, as they exert no significant mechanical constraint on the crown. Their role is purely passive, providing contact support to simulate the real assembly on the aircraft. As shown in Figure 24, they are modelled as eight fixed support points located at the counterbored holes of the crown.

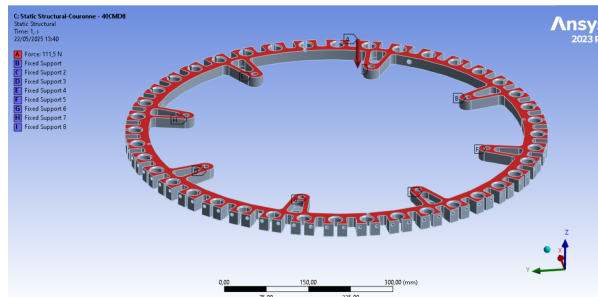


Figure 24: Boundary Conditions - 8 Springs

Conversely, in the extreme configuration, the crown is assumed to rest on three rigid supports. As illustrated in Figure 25, they are modelled as three fixed points placed 120° apart, simulating a rigid tripod ensuring stable, undeformable contact.

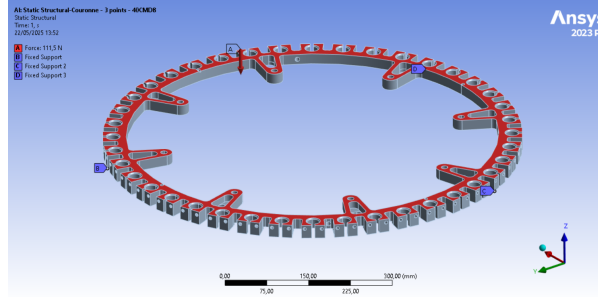


Figure 25: Boundary Conditions - 3 Rigid Supports

The resulting deformations are shown in Figure 26.

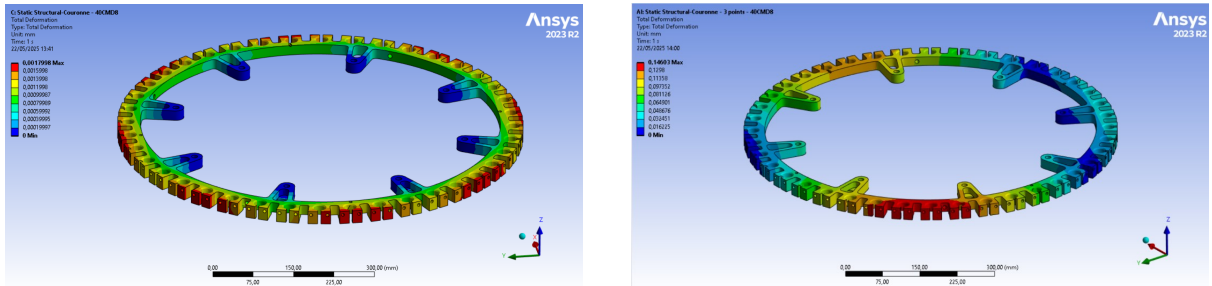


Figure 26: Deformations for Both Hypotheses

Table 5 compares the simulated flatness values with the actual measured deformations for both configurations.

Table 3: Measured vs Simulated Flatness - 2 Hypotheses

Configuration	Measured Flatness	Simulated Flatness
Crown placed on 8 springs	0.162 mm	0.0018 mm
Crown placed on 3 rigid supports	0.250 mm	0.146 mm

The results show that the eight-spring hypothesis leads to a theoretical deformation in the micron range, while the actual flatness deviation is 0.162 mm, which represents a very significant deviation. In contrast, the three-support hypothesis yields a simulated flatness of 0.146 mm, much closer to the measured 0.250 mm, with a gap of approximately 0.1 mm.

These observations are consistent with the findings of Lee [21], who highlights that boundary conditions play a critical role in finite element simulations. In the present case, since the material properties and the applied load are identical across both models, the variation in results is clearly attributable to the type of support conditions used.

However, Lee’s work remains limited to idealised contact modelling and does not explore the more complex behaviour of soft or elastic supports such as springs. This limitation represents one of the key gaps identified in the literature review and directly impacted the modelling choices in this study.

As a result, the hypothesis based on eight undeformable springs proved difficult to translate into a realistic simulation model. The simplification led to an overconstrained system that fails to reproduce the physical deformation observed experimentally.

It can therefore be concluded that the first hypothesis is not suitable for accurately representing the mechanical behaviour of the crown. This reinforces the difficulty of numerically modelling soft support conditions, a major gap that was previously identified in the literature.

On the other hand, the three-support hypothesis provides more representative results and will be retained for the remainder of the study. To better approximate real behaviour, a correction factor of 60% ($0.15 / 0.25$) will be applied to all subsequent simulated results.

4.3.2 Mesh Convergence Study

The finite element method inherently requires a mesh sensitivity analysis to ensure the reliability of simulation results. In this study, three mesh configurations were tested in order to determine the best compromise between simulation accuracy and computational cost. Table 4 summarises the characteristics and results of each mesh, while Figure 27 illustrates the convergence trend.

Table 4: Mesh Sensitivity Study

Mesh Type	No. of Elements	Average Element Size	Max Displacement	Computation Time
Coarse	32 687	10 mm	0.152 mm	11 s
Medium	130 261	5 mm	0.146 mm	71 s
Fine	516 391	3 mm	0.141 mm	419 s

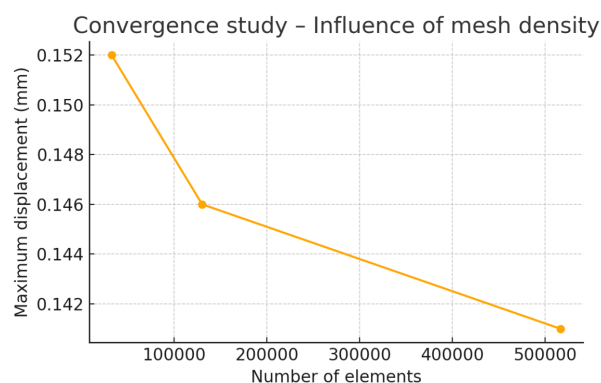


Figure 27: Convergence Curve

The coarse mesh, with an average element size of 10 mm, yielded a displacement of 0.152 mm in just 11 seconds. In contrast, the fine mesh reduced the displacement to 0.141 mm, but required over 400 seconds to compute. The medium mesh offered an intermediate result (0.146 mm) for a significantly shorter computation time of 71 seconds.

The convergence curve clearly shows that beyond the medium mesh, the gains in accuracy become marginal, despite a considerable increase in number of elements.

These findings are consistent with the work of Hemesh [24], who demonstrated that increased mesh density improves result stability and precision (in terms of stress and deformation), though at a higher computational cost.

However, while Hemesh's study is valuable, it does not address adaptive meshing strategies, which could have offered a more efficient alternative by refining only the high-strain areas. Although such an approach was not implemented here, it represents a potential direction for future work.

Discussion and mesh selection

It is worth noting that ANSYS was unable to solve the model for element sizes smaller than 3 mm due to memory limitations. The solver could not assemble the system matrices, which imposed a practical lower bound for the mesh refinement.

Given the iterative nature of the design process in this study, computation time is a critical factor. The medium mesh provides a significant improvement over the coarse mesh (-0.006 mm) while keeping a reasonable simulation time. In contrast, the fine mesh only yields a negligible additional gain (-0.005 mm) at a computational cost six times higher.

For these reasons, the medium mesh is retained for all subsequent simulations as the optimal compromise between accuracy and computational efficiency.

With a validated ANSYS model that reliably reproduces the actual deformation behaviour, the study can now shift focus toward identifying and evaluating potential design improvements.

4.4 Analyse

Given the constraints of industrial prototyping, it is neither realistic nor cost-effective to manufacture and test multiple crown designs through physical trials alone. Therefore, finite element simulations using ANSYS were deemed essential for evaluating design alternatives in a predictive and non-destructive manner. As noted by Lee [21], such tools enhance the credibility of the design process by allowing mechanical behaviours, such as stress distribution and deformation, to be assessed before fabrication.

In line with the iterative approach defined in the Methods section, the design options were progressively refined based on a set of intermediate research questions. These questions served as decision filters, guided by performance indicators such as extrapolated deformation and component mass. The analysis followed a funnel-shaped logic: broad design hypotheses were first explored, then narrowed down using individual performance criteria, before finally being cross-compared to identify the most balanced and robust design solution.

Broad Design

This initial stage explores a range of potential improvement strategies without yet applying performance indicators as selection filters. This open-ended exploration phase is common in engineering projects, where creativity and innovation are encouraged (provided they remain grounded in technical and industrial feasibility). Such an approach aligns well with the applied nature of this research.

At the end of the Define phase, it was concluded that the flatness issue could originate from both internal and external factors affecting the crown. Therefore, both aspects are investigated in this section.

4.4.1 Clamping

Beyond the previously discussed uneven spring heights, the applied tightening torque on the 20 clamping brackets may also induce deformation and compromise flatness.

The current torque value, set at 3 Nm at the client's request, needed to be converted into an equivalent force, as ANSYS only handles force inputs.

The CAD model of the STHC M8x20 pressure screw used is shown in Figure 28.

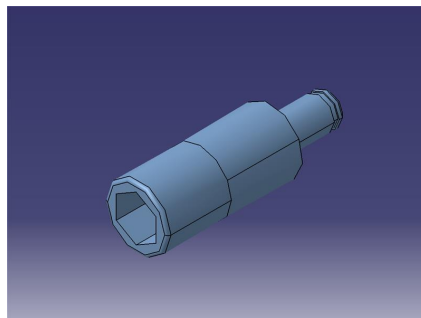


Figure 28: Pressure Screw

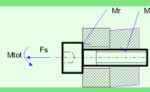
Its main characteristics are as follows:

- **Nominal screw diameter (M8):** 8 mm
- **Thread pitch:** 1.25 mm (source: Guide du dessinateur industriel, Chevalier A. [39])
- **Mean bearing surface diameter:** 6 mm
- **Friction coefficient (thread/bearing):** 0.16
- **Applied torque:** 3 Nm

These values were used as input parameters in an online torque-to-force calculator [40], the interface of which is shown in Figure 29.

Détermination des forces de serrage en présence dans un assemblage vissé.

Croquis



Equations

$$\tan \alpha = \frac{P}{d_{vis}} \quad [\mu = \tan \alpha]$$

$$M_r = \frac{1}{2} d_{appui} F_s \mu_{appui} \quad M_s = F_s \tan(\alpha \pm \rho_v) \frac{d_{vis}}{2}$$

$$M_{tot} = M_r + M_s$$

Calcul

Données

Diamètre de filet	$d_{vis} =$	0,008	[m]
Pas du filet	$P_{vis} =$	0,00125	[m]
Diamètre moyen de la face d'appui	$d_{appui} =$	0,006	[m]
Coefficient de frottement dans le filet	$\mu_{vis} =$	0,16	[-] v
Coefficient de frottement dans l'appui	$\mu_{appui} =$	0,16	[-] v
Couple de serrage	$M_{tot} =$	3	[Nm] v

Lancer le calcul

Résultats

Angle du filet	$\alpha =$	2,84	[°]
Angle de frottement dans le filet	$\rho_v =$	9,09	[°]
Moment de frottement dans l'appui	$M_r =$	1,08	[Nm]
Moment de torsion dans la vis (au serrage)	$M_s =$	1,91	[Nm]
Couple de desserrage (théorique...)	$M_{dev} =$	0,09	[Nm]
Force de serrage	$F_s =$	2263	[N]

Figure 29: Online Torque-to-Force Calculator

The resulting clamping force for a 3 Nm torque is 2 263 N, equivalent to 226.3 kg. Considering that the vanes can be deformed manually, this value appeared excessive.

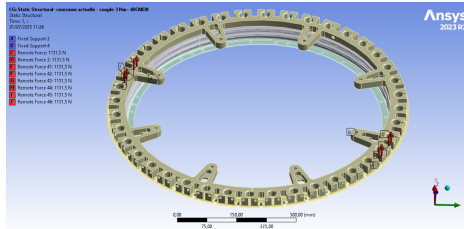
A second scenario was thus tested with a reduced torque of 1 Nm. Using the same calculator, the corresponding force was 754 N (75.4 kg), which is approximately one-third of the initial value.

As illustrated in Figure 30, this clamping force is distributed across two surfaces via dual pads (highlighted in red). Hence, each contact zone was assigned half the total force input in the simulation: 1 131.5 N for the 3 Nm case and 377 N for the 1 Nm case.

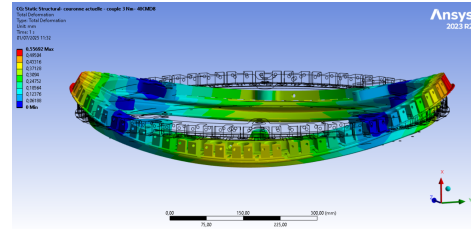


Figure 30: Clamping Forces Areas

For the ANSYS simulation, the most unfavourable case was considered: two pairs of vanes located 180° apart were assumed to present flatness issues corrected by clamping. The initial hypothesis of eight spring points was excluded previously due to its impracticability. The model therefore considered only two fixation zones located at opposite spring counterbores. Clamping was applied through two sets of brackets placed at 180°, using all 40 brackets available instead of just 20. Figure 31 presents the boundary conditions reflecting this hypothesis and the amplified deformation shape.



(a) Boundary conditions



(b) Amplified deformation shape

Figure 31: Current Crown - 3 Nm

The amplified results clearly show a U-shaped deformation, indicating a symmetrical tendency caused by clamping loads.

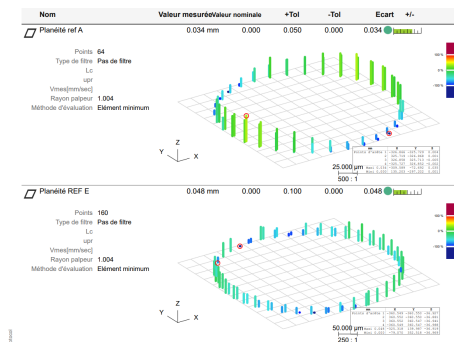
The table 5 below summarises the extrapolated deformation values obtained in both scenarios:

Table 5: Clamping Torque - 3 Nm vs 1 Nm

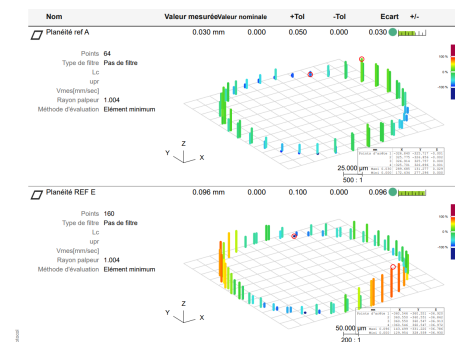
3 Nm	1 Nm
0.891 mm	0.236 mm

The 3 Nm torque generated a deformation of 0.89 mm, while the 1 Nm torque produced a deformation of only 0.24 mm, a factor of nearly 3.

Additionally, physical tests were conducted on site. The sealing flange support was clamped first at 1 Nm, then at 3 Nm, and its flatness was measured on a granite surface after each operation. The test results are shown in Figure 32.



(a) Clamping torque – 1 Nm



(b) Clamping torque – 3 Nm

Figure 32: Experimental Validation - Clamping Torque

With:

- **Ref A:** Lower surface of the sealing flange in contact with the fixture base
- **Ref E:** Vane surface where clamping is applied

The results confirmed that a 3 Nm torque induced a flatness deviation twice as large as that caused by a 1 Nm torque: 0.096 mm versus 0.048 mm on reference plane E.

Both ANSYS simulations and experimental data led to the same conclusion: the higher the torque, the greater the induced deformation (by a factor of 2 to 3). This supports the hypothesis that clamping torque has a direct impact on the crown's and component's flatness.

These results are consistent with the principles outlined by CarrLane [11], which emphasise the risks of excessive or poorly distributed clamping forces in inducing deformation, particularly when parts are overconstrained. While the 3-2-1 principle ensures geometric stability, it does not provide guidance on torque sensitivity or elastic behaviour in compliant components. The simulations and physical tests carried out in this study confirm that a tightening torque of 3 Nm leads to a two to threefold of the deformation of the 1 Nm torque, validating the impact of clamping load on flatness.

In parallel, these observations echo the findings of Shi et al. [13], who demonstrate that deformation in flexible components can be significantly reduced by adapting the fixture's compliance. Although their work focuses on grinding applications and makes use of advanced adaptive systems, the underlying principle (adjusting clamping to the mechanical response of the part) remains fully applicable to the present case.

Overall, the correlation between torque and deformation observed in both numerical and experimental results reinforces the need for fixture designs that balance mechanical constraint and local flexibility, particularly when dealing with geometries highly sensitive to clamping-induced stress. However, it is worth noting that neither of these sources specifically addresses ultra-low deformation scenarios, which remain a critical gap in current research

Moreover, the clamping sequence was found to influence the resulting deformation. Until now, the brackets had been tightened sequentially in a clockwise direction. However, this method may generate residual stresses. To address this, a cross-pattern tightening sequence was proposed to balance these stresses.

Workshop testing confirmed this hypothesis: the new cross-pattern significantly reduced flatness deviation.

In summary, a clamping torque of 1 Nm was found to be sufficient. This recommendation was presented to the client to encourage further discussion, without challenging the 3 Nm currently in use or imposing a change. In parallel, the clamping procedure was revised to adopt a cross-pattern tightening sequence instead of the previous clockwise approach.

Now that external loading effects have been addressed, the next step focuses on optimising the crown’s internal geometry.

4.4.2 Material

Before modifying the geometry of the crown, it was essential to assess whether a change of material could provide increased stiffness. As is well known, the mechanical behaviour of a component is strongly influenced by its material properties, particularly Young’s modulus, which quantifies resistance to elastic deformation. The higher the modulus, the more rigid the material.

The current material used is 40CMD8 (also referred to as P20 in the AISI standards), a tool steel commonly employed for moulds and fixtures. Its mechanical properties, already used in the simulation assumptions in the Measure section, are as follows:

- Young’s modulus: 205 000 MPa
- Density: 7 850 kg/m³
- Poisson’s ratio: 0.3

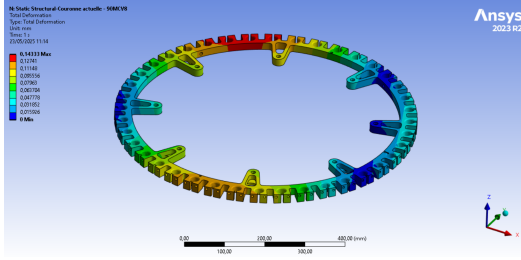
To investigate potential alternatives, four other tool steels were selected for comparison. Their relevant properties are summarised in Table 6.

Table 6: Mechanical Properties of Selected Tool Steels

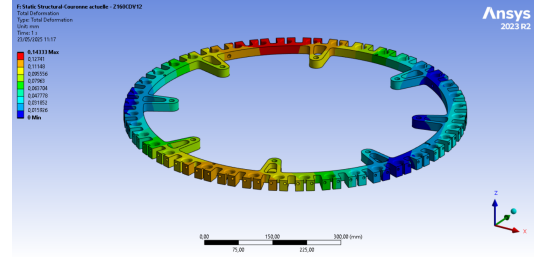
Material	Density	Young’s Modulus	Poisson’s Ratio
90MCV8	7.85 kg/dm ³	210 GPa	0.3
Z160CDV12	7.85 kg/dm ³	210 GPa	0.3
Z200C12	7.70 kg/dm ³	200 GPa	0.3
55NCDV7	7.80 kg/dm ³	215 GPa	0.3

All steels (except Z200C12) exhibit a higher Young’s modulus than the current material, indicating a potentially greater stiffness without a significant increase in mass, as the densities remain comparable. Z200C12 was still included in the analysis due to its lower density.

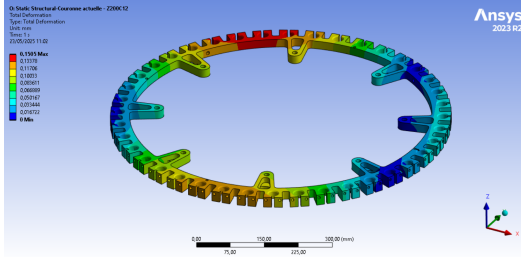
The corresponding ANSYS simulations were adjusted for each material’s density to ensure consistent loading based on self-weight. The deformation results for each case are shown in Figure 33. The previously validated assumption of three rigid support points was used for all simulations and will be retained throughout the remainder of this study.



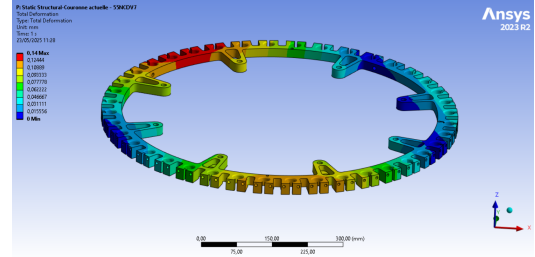
(a) 90MCV8



(b) Z160CDV12



(c) Z200C12



(d) 55NCDV7

Figure 33: Materials Comparison - ANSYS

Table 7 presents the extrapolated deformations for each material, assuming an identical crown geometry. This approach enables a direct comparison of material influence on structural stiffness.

Table 7: Material - Extrapolated Deformation

Material	Extrapolated Deformation
40CMD8 (current)	0.25 mm
90MCV8	0.229 mm
Z160CDV12	0.229 mm
Z200C12	0.25 mm
55NCDV7	0.224 mm

All tested materials yielded similar deformation results, ranging from 0.224 mm to 0.25 mm. The differences are negligible in practical terms. This confirms that simply changing the material does not lead to a significant gain in stiffness for this application.

These results are consistent with the observations of Robert [27], who noted that tool steels are generally selected for their high rigidity and dimensional stability, making them well suited for fixture applications. Indeed, the simulated deformations remain relatively low for standard, rigidly constrained fixtures.

In addition, as highlighted by Mantle [28], P20 (40CMD8) offers excellent machinability without requiring heat treatment, making it a practical and cost-effective choice. This justifies its original selection for the current design.

However, it was surprising to observe such a small variation between materials with different elastic moduli. This points again to a key limitation in the existing literature: most studies do not address ultra-low deformation scenarios or the interaction with compliant supports such as springs.

In conclusion, this investigation confirmed that changing the crown’s material does not offer meaningful improvement in stiffness. The remainder of this study will therefore focus on modifying the geometry of the crown, while retaining the original 40CMD8 material.

4.4.3 Addition of Geometric Features

A well-known method in the industry to increase the stiffness of a component is the addition of geometric features. Several types exist, and I have chosen to investigate the following: ribs, stiffeners and extra thickness.

As shown by Lin [16], topology optimisation is also a powerful method to reinforce parts while minimising mass. It is commonly applied in motorsport, however, it is not suitable for dimensional inspection tooling.

To model the new crown designs, I chose to use the 3D design software CATIA V5. Beyond my existing experience with this tool, particularly through my Master’s programme at Cranfield, additional sources confirm the relevance of this software. Dr. Kovacevic [17], for example, highlights in his academic materials the extensive capabilities offered by CATIA, which are well suited to this case, such as part design and assembly. However, as stated in my literature review, these sources remain general and do not present specific industrial application cases.

Moreover, CATIA proved particularly useful in my iterative design process. As presented by Wang [19], parametric modelling enables efficient changes to key parameters while automatically updating the rest of the geometry. This capability is crucial in my case, where several versions needed to be tested by varying parameters such as thickness. Figure 34 illustrates the use of parametric modelling on the thickness dimension via a functional parameter.

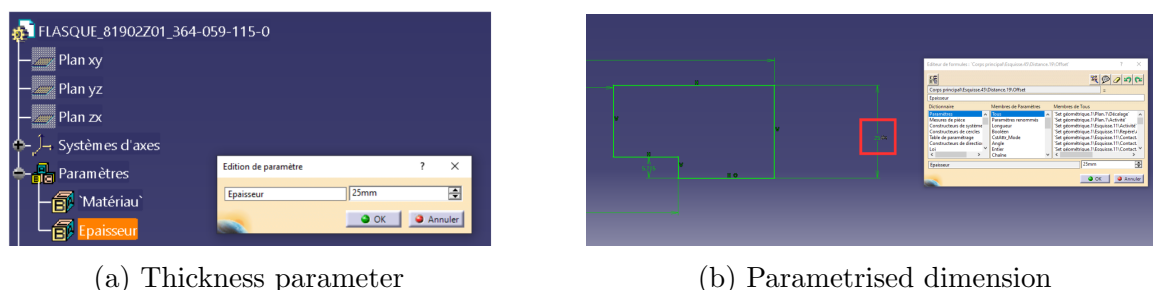


Figure 34: Parametric Design of the Crown Thickness

After several iterations, three design alternatives clearly stood out as the most relevant for comparison:

- A rib with a height of 7.5 mm positioned across the crown width
- Star-shaped stiffeners placed at the eight fixing arms, having the same width
- A 7 mm extra thickness of the base crown, increasing the total thickness from 25 mm to 32 mm

Figure 35 presents the 3D models and corresponding ANSYS simulations for these three geometric additions. For visual clarity, the added features are shown in red, while the original crown remains turquoise.

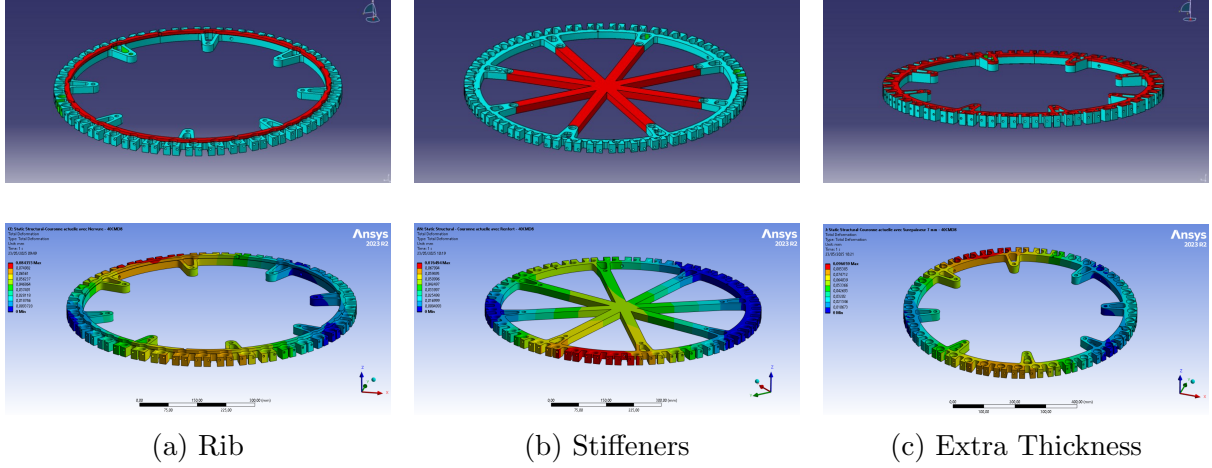


Figure 35: Geometric Additions - CATIA & ANSYS

Table 8 summarises the different improvement variants tested on the current crown design. Each modification is presented along with the resulting total mass (initial mass of 11.37 kg plus the added geometry) and the extrapolated deformation at 60%.

As a reminder, the current crown features a thickness of 25 mm, a mass of 11.37 kg, and a measured deformation of 0.25 mm.

Table 8: Comparison of Geometric Additions

Type	Total Mass	Extrapolated Deformation
Rib	13 kg	0.134 mm
Stiffeners	20 kg	0.122 mm
Extra Thickness	15 kg	0.154 mm

All proposed variants provide improved stiffness and reduced deformation compared to the original crown. The rib configuration offers the lowest mass (13 kg), with moderate stiffness improvement. The stiffeners design provides the lowest deformation (0.122 mm) but results in the highest mass (20 kg). The extra thickness option offers an intermediate mass (15 kg) but the least favourable stiffness gain.

To visually summarise the trade-offs between performance indicators (mass, stiffness, and thickness), a radar diagram is presented in Figure 36.

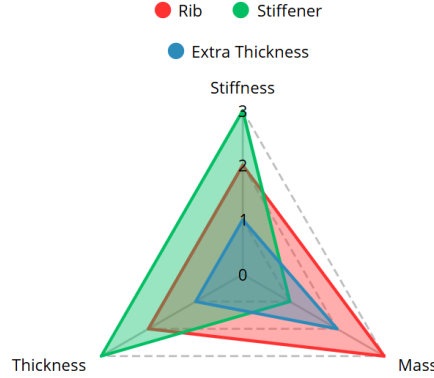


Figure 36: Radar Diagram of Geometric Additions

A cross-analysis of the performance indicators highlights that the stiffener solution offers the best rigidity without increasing the original thickness (25 mm). Despite a significant increase in mass, it remains the preferred choice.

These results are generally aligned with the conclusions of Lin [16], confirming that geometric additions enhance stiffness. However, a point of divergence arises, as Lin claims that these modifications do not affect the mass, which contradicts my results that show a clear increase in total weight.

The stiffener solution is therefore selected for further analysis, as the primary objective is to maximise crown rigidity in order to meet flatness tolerance requirements. Weight optimisation may be considered at a later stage.

Now, all potential avenues have been explored, and it has been established that:

- Applied torque influences deformation $\rightarrow$ 1 Nm retained
- Material choice has negligible impact $\rightarrow$ 40CMD8 maintained
- Stiffeners offer the best rigidity performance

It is now time to refine the study with question-driven filters, as introduced earlier.

Research Question 1: How can deformation be reduced ?

4.4.4 Types of Stiffeners

As required by the first performance filter, the initial focus is placed on reducing deformation, which directly translates into increasing structural stiffness.

Previous results demonstrated that adding stiffeners provided the most effective improvement in rigidity. Therefore, this section evaluates and compares several stiffener designs to identify the most effective configuration.

Following the same logic applied during the geometric enhancement study, the iterative modelling process led to the development of numerous variants. To ensure clarity and avoid overloading the report, only the three most promising configurations were selected for detailed analysis. This pre-selection acts as an implicit filter, allowing focus on the most relevant outcomes.

The three retained designs are:

- Stiffeners
- Stiffeners + Circle
- Stiffeners + Internal crown

Figure 37 illustrates both the CATIA 3D models and the corresponding ANSYS simulation results for these three variants.

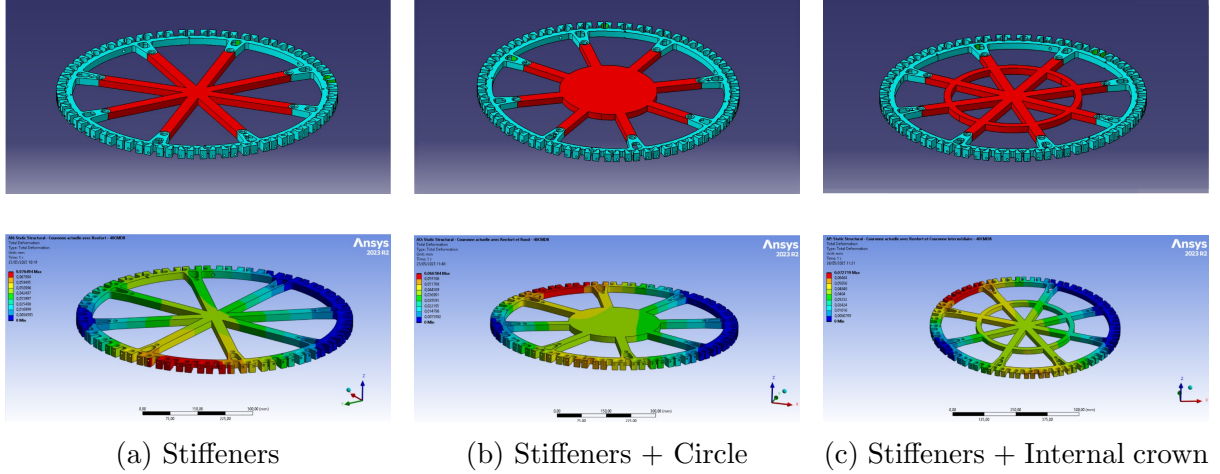


Figure 37: Retained Stiffener Designs - CATIA & ANSYS

Table 9 presents, for each configuration, the total mass and the extrapolated deformation, assuming a constant crown thickness of 25 mm.

Table 9: Performance Comparison of the Three Stiffener Designs

Stiffener Design	Total Mass	Extrapolated Deformation
Stiffeners	20.21 kg	0.122 mm
Stiffeners + Circle	25.16 kg	0.107 mm
Stiffeners + Internal crown	23.01 kg	0.116 mm

At constant thickness, the configuration with stiffeners results in the lowest mass (20.21 kg) but also exhibits the highest deformation. The addition of a circle increases the mass by approximately 5 kg, but achieves the best rigidity performance. The third configuration, involving an internal crown, yields intermediate results in terms of both mass and deformation.

To further investigate the role of geometry, a 30 mm-thick version of the “Stiffeners” model was simulated. The total mass increased from 20.21 kg to 25.34 kg, and the extrapolated deformation improved from 0.122 mm to 0.108 mm.

These results suggest that the geometry of the stiffeners alone does not directly dictate performance. Rather, mass is the primary driver of rigidity in this context.

This finding leads to the second performance filter, defined in the Methods chapter, which focuses on mass reduction.

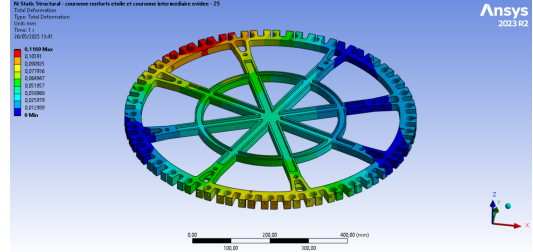
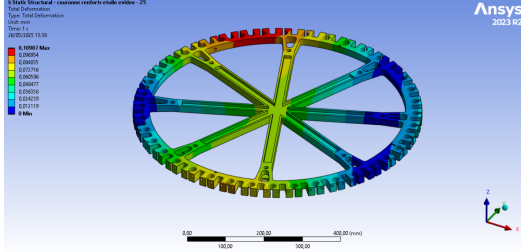
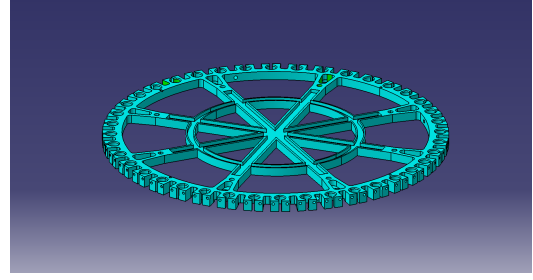
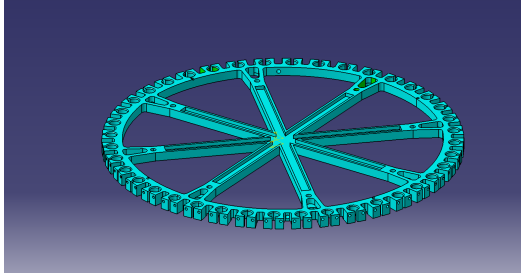
Research Question 2: How can mass be reduced ?

4.4.5 Hollowed Stiffeners

Having established that deformation is directly dependent on mass, this parameter is now formally introduced as a new performance criterion to minimise.

In the context of structural stiffeners and manufacturing, one of the most common approaches to reduce weight is to hollow out the beams (transforming them from solid to tubular or shell-like geometries). This strategy aims to optimise the stiffness-to-mass ratio.

As the previously studied configuration ”stiffeners + circle” was found unsuitable, it is excluded from this phase of the study. The 3D models and ANSYS simulations for the two remaining hollowed configurations are shown in Figure 38.



(a) Hollowed stiffeners

(b) Hollowed stiffeners + Internal crown

Figure 38: Hollowed Stiffener Designs - CATIA & ANSYS

In order to assess and compare both hollowed configurations, the thickness was varied to generate different mass and deformation values. These are reported in Table 10. The upper limit is constrained by a maximum allowable crown weight of about 30 kg, previously defined based on the capacity of the rotary table on which the fixture is mounted.

Table 10: Comparison of the Two Hollowed Stiffener Configurations

	Hollowed stiffeners		Hollowed stiffeners + Int. crown	
Thickness	Mass	Deformation	Mass	Deformation
25 mm	15.42 kg	0.174 mm	16.46 kg	0.185 mm
30 mm	18.87 kg	0.123 mm	18.83 kg	0.133 mm
35 mm	22.33 kg	0.0912 mm	22.16 kg	0.101 mm
40 mm	25.78 kg	0.0704 mm	25.49 kg	0.0784 mm
45 mm	29.24 kg	0.0563 mm	28.82 kg	0.0654 mm

The extrapolated deformation initially measured at 0.25 mm (on site, for a 25 mm thick, 11.37 kg crown supported by three pins) is significantly reduced across all configurations, ranging between 0.185 mm and 0.056 mm depending on thickness. As expected, increasing the thickness leads to higher mass and improved stiffness. Nevertheless, even the largest tested configuration remains within a realistic range (maximum 45 mm).

To clearly visualise the mass-to-deformation relationship, a comparison plot was generated (Figure 39). It contrasts the current crown with both hollowed configurations.

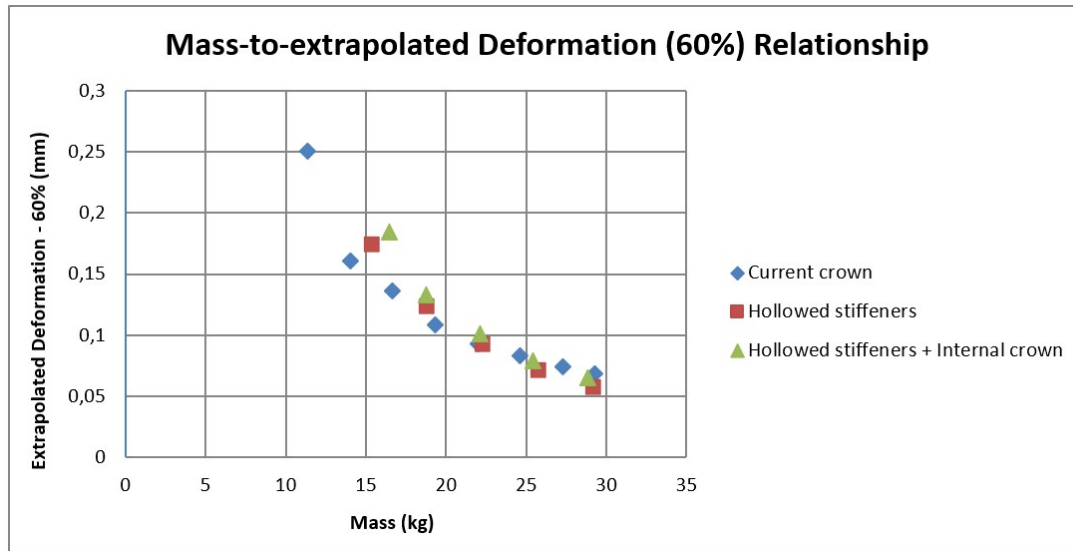


Figure 39: Mass vs Extrapolated Deformation (60%)

The graph confirms a clear trend: deformation decreases as mass increases. Prior to this study, it could be hypothesised that beyond a certain point, adding weight would yield negligible gains in stiffness. However, in this case, due to the very tight flatness tolerance, every kilogram plays a significant role.

These results support the choice of the simpler hollowed stiffener configuration. Since its extrapolated deformation is similar to that of the more complex design (with intermediate crown), the latter is considered unnecessary. Moreover, retaining a more open structure is beneficial to ensure access to probing points during inspection.

To conclude this second research question, it is essential to introduce an additional notion: manufacturability. One of the key values of this study lies in its strong connection to real industrial constraints. The analysis combines theoretical modelling with actual physical tests, offering grounded and applicable insights.

A common pitfall among engineering students is the development of theoretically sound solutions that prove infeasible to manufacture using the processes or resources actually available. Ensuring that the selected design remains compatible with existing production capabilities is therefore a critical requirement.

Research Question 3: How can the selected solution be made manufacturable ?

4.4.6 Selected Solutions

While hollowed configurations offer a promising stiffness-to-mass ratio, they entail increased manufacturing complexity. Particular care must be taken to anticipate and mitigate residual stresses caused by machining operations, especially within internal cavities [31]. Although Weber’s work was not focused on tooling materials, the conclusions regarding stress-related deformations remain applicable. Soori et al. further emphasise that material properties, cutting conditions, and cooling strategies all interact to influence the internal stress profile within machined parts [30]. If uncontrolled, such residual stresses can result in permanent warping and compromise fixture flatness, particularly critical in high-precision dimensional inspection contexts.

Based on the conclusions previously established, the final parameters selected include the use of 40CMD8 steel, solid stiffeners, and a maximum thickness of 45 mm.

To address the initial requirement (flatness < 0.025 mm), two complementary solutions are proposed: a short-term solution based on the current crown, allowing continued inspection activities, and a long-term solution involving the design of a new crown, providing a more sustainable resolution to the issue.

Short-term Solution

Previous analyses revealed discrepancies in spring height, which justifies the implementation of a system for more precise adjustment. The simplest and most well-documented solution involves the use of precision shims.

The principle consists in using eight shims of 39.59 mm height (i.e. 0.02 mm above the nominal dimension) to support the crown. Once in place, the springs are brought into contact with the counterbored holes. The shims are then removed, and flatness is verified using feeler gauges. Figure 40 illustrates one shim in position.

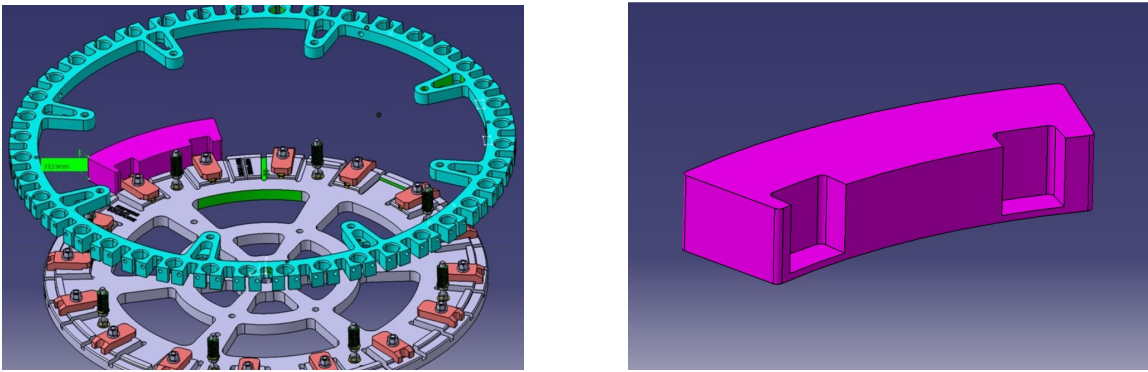


Figure 40: Spring Height Adjustment System

This methodology appears to enable the reuse of the current crown in a compliant and functional way.

Long-term Solution

The final version of the new crown features star-shaped stiffeners with a thickness of 45 mm. To eliminate the risk of residual stresses caused by machining, the hollowed design was abandoned in favor of thinner, solid ones. This design is inspired by civil engineering practices, where I-beams provide superior flexural stiffness when oriented on edge.

Using thinner stiffeners also helps reduce the overall mass and allows for the removal of the eight original “ears”, enabling lateral rather than vertical probing. Naturally, the eight counterbored holes are retained, even in the absence of the original ears. Figure 41 presents the 3D model and ANSYS simulation results.

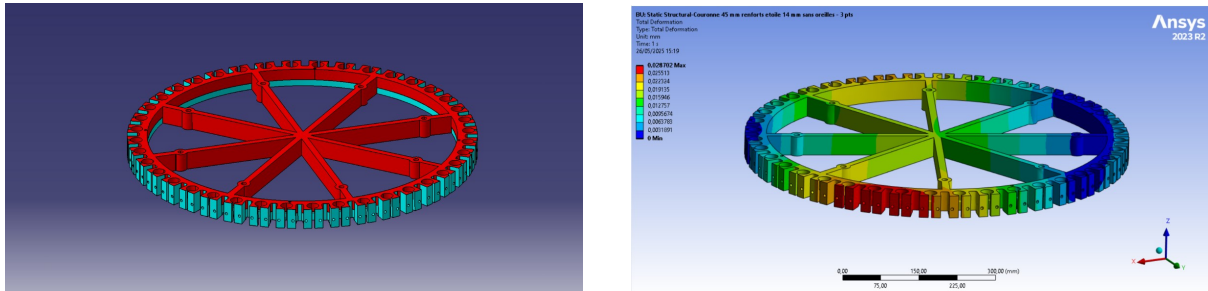


Figure 41: New Crown - CATIA & ANSYS

Table 11 presents the total mass and extrapolated deformation (at 60%) for this new design, enabling a performance comparison with previous configurations.

Table 11: Thin Stiffeners with 45 mm Thickness

Mass	Extrapolated deformation
29.8 kg	0.0459 mm

The resulting deformation of 0.046 mm represents the best performance achieved so far, and it is obtained while staying within the maximum allowable mass. This validates the effectiveness of this configuration in terms of stiffness.

Summary

These results confirm that the selected solutions offer a significant improvement in both stiffness and functional usability. The shim-based system ensures immediate reusability of the current crown, while the new reinforced design achieves over 80% deformation reduction compared to the original configuration, without exceeding the allowable mass.

These options must now be assessed against industrial constraints such as standards, ergonomics, and manufacturability, as well as compliance with operational requirements. In line with the DMAIC methodology, the study now shifts from the analytical stages (Define, Measure, Analyse) to the decision-making phases (Improve, Control), aiming to validate and refine the most effective solution for industrial deployment. Both short and long-term solutions will therefore be developed further to ensure full alignment with manufacturing, ergonomic, and operational constraints.

4.5 Improve

With both solutions now defined and selected, the next step involves refining their implementation by incorporating additional factors that, while previously non-prioritised, are essential for detailed design and industrial deployment.

Short-term Solution

The production of the eight shims will be carried out internally at the Portugal site. A manufacturing drawing has therefore been prepared, as shown in Figure 42.

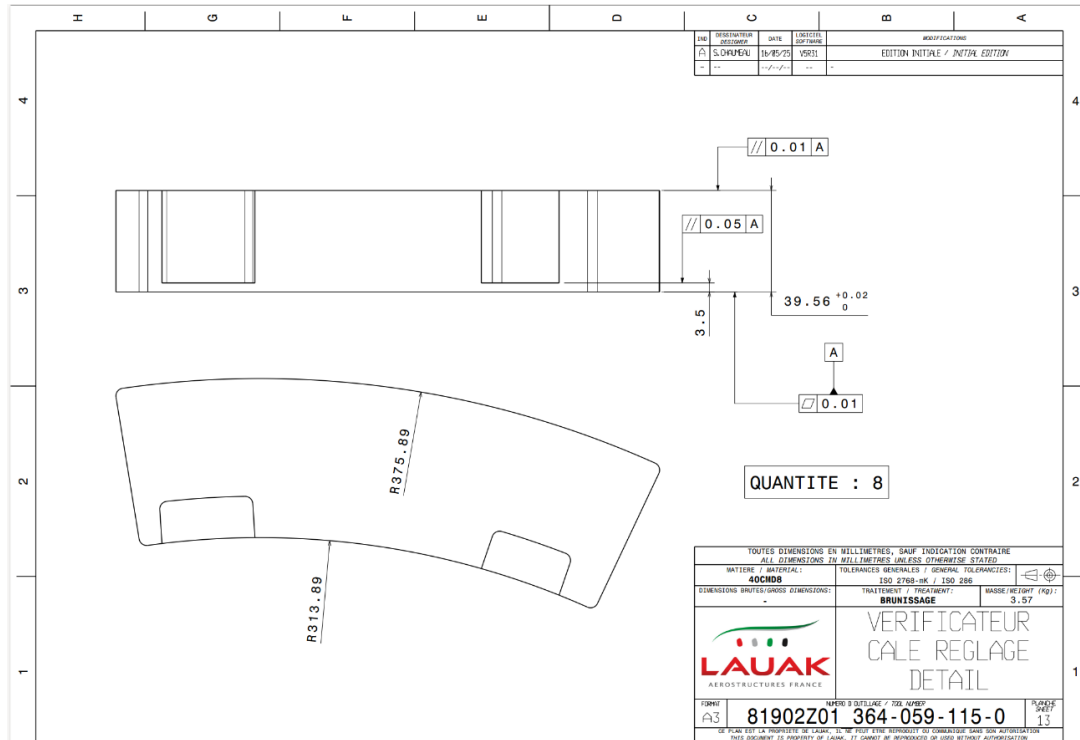


Figure 42: Manufacturing Drawing - Shims

As previously mentioned, the use of a cross-tightening sequence combined with a torque of 1 Nm has proven effective during physical testing, aligning well with theoretical predictions. These parameters are therefore confirmed and retained for implementation.

The short-term solution thus comprises:

- Eight precision shims for spring height adjustment
- Cross-tightening sequence
- Torque setting of 1 Nm

Long-term Solution

From a manufacturability perspective, the final crown design must be revised to account for several technical constraints. These are summarised in Table 12, which also includes the corresponding design adaptations.

Table 12: Integration of Technical, Financial, and Social Constraints

Type	Description	Proposed Adaptations
Technical	Machining and access to probing points	- Central hole ($\varnothing 25$ mm) added to define the probing program origin - Reinforcement around vertical holes located too close to the edge - Fillets added ($R \geq 5$ mm) on sharp edges to improve machinability
Financial	Avoid modifying the probing program or purchasing new equipment	- Retention of the existing probing program - Use of a longer-stroke probe as an alternative to replacing the current one
Social	Operator health and safety (ergonomics)	- Four vertical M10 threaded holes added to replace the original radial ones for improved lifting - Compliance with standard NF X35-109 (AFNOR): mandatory handling assistance for loads > 25 kg - Lifting system integrated in line with LAUAK's CSR commitments

Solutions have been found for all the identified constraints, despite their differing nature. The CAD model was subsequently updated in CATIA to reflect these changes, as illustrated in Figure 43. For clarity, the added features are shown in orange.

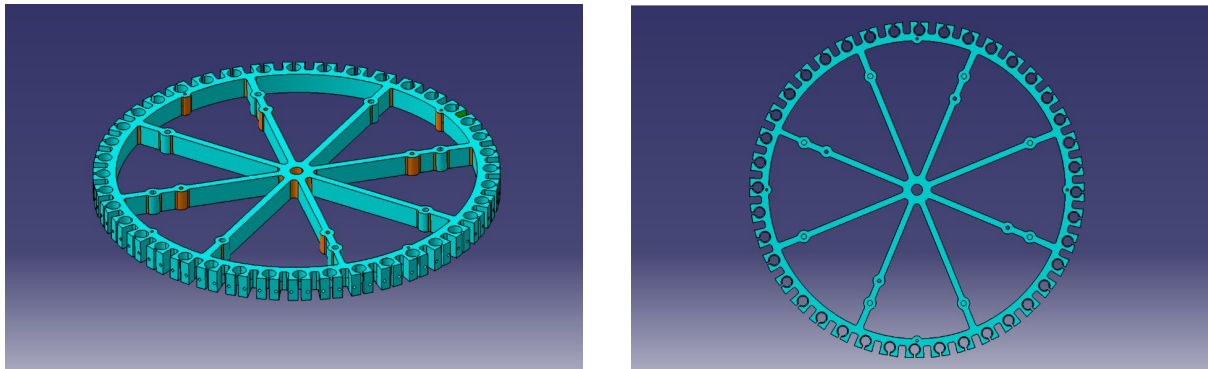


Figure 43: New Crown Geometry - CATIA

The final geometry showed minimal variation throughout the process, indicating that the preliminary design phase had been carefully considered. The research-question-driven methodology provided a clear and effective framework, guiding the development toward a stable and consistent solution.

In parallel, two studies were carried out to optimise the remaining performance indicators: deformation (extrapolated at 60%) and total mass. The first investigates the

self-weight behaviour of the updated design, while the second analyses its behaviour when clamped to the sealing flange support.

Analysis of the Crown Under Self-Weight

The earlier simulations demonstrated that the 45 mm thick crown (29.8 kg) provides optimal rigidity. However, it is now relevant to investigate whether a lower-mass alternative could still satisfy the stiffness requirements. Figure 44 presents the ANSYS simulation results for the updated model.

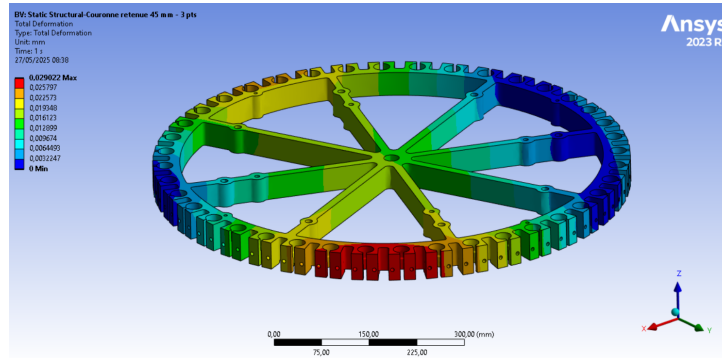


Figure 44: New Crown - ANSYS

The corresponding results, including extrapolated deformation and mass for each thickness, are summarised in Table 13.

Table 13: Thickness Study - New Crown

Thickness	Mass	Extrapolated Deformation
30 mm	19.25 kg	0.090 mm
35 mm	22.89 kg	0.068 mm
40 mm	26.56 kg	0.053 mm
45 mm	30.2 kg	0.046 mm

The integration of the previously defined technical constraints resulted in a slight increase in total mass, marginally exceeding the 30 kg target. Nevertheless, due to the limited deviation and the significant gain in rigidity, the 45 mm configuration cannot be ruled out. As observed earlier, an increase in mass generally correlates with a decrease in deformation. This is directly linked to the thickness of the part: the greater the thickness, the higher the resulting mass. Finally, the 40 mm variant remains a viable alternative, offering a favourable balance between mass and stiffness.

Following a technical meeting, the 45 mm design was unanimously approved for the following reasons:

- Increased thickness allows for potential future re-machining if necessary.
- A thicker geometry ensures greater probing margin in case a new stylus with extended stroke must be acquired.

Analysis of Clamping Torque - 1 Nm vs 3 Nm

As previously mentioned, the clamping torque currently used is 3 Nm. However, both simulations and on-site tests suggest that reducing it to 1 Nm would significantly lower deformation while maintaining sufficient contact. A simulation was therefore performed in ANSYS to assess the impact of both torque levels on the assembled structure.

The updated assembly including the crown and sealing flange support was modelled in CATIA, as shown in Figure 45. This geometry served as the basis for the numerical simulations.

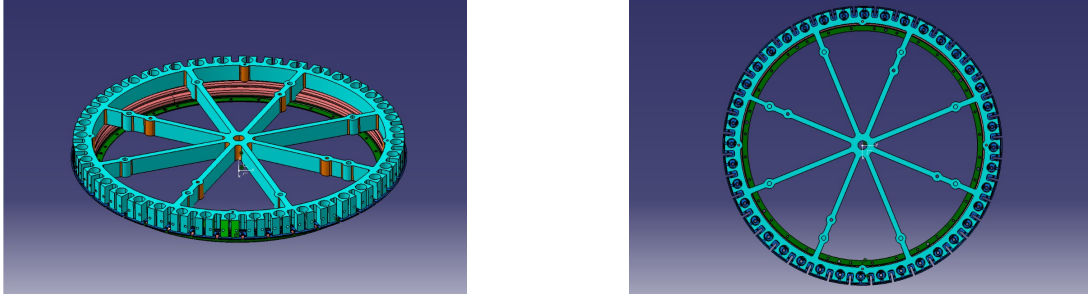


Figure 45: New Crown with Sealing Flange Support

As before, the simulated tightening setup reproduces a worst-case scenario, in which two pairs of clamps (180° apart) engage all 40 available brackets, rather than the usual 20. Figure 46 shows the applied boundary conditions.

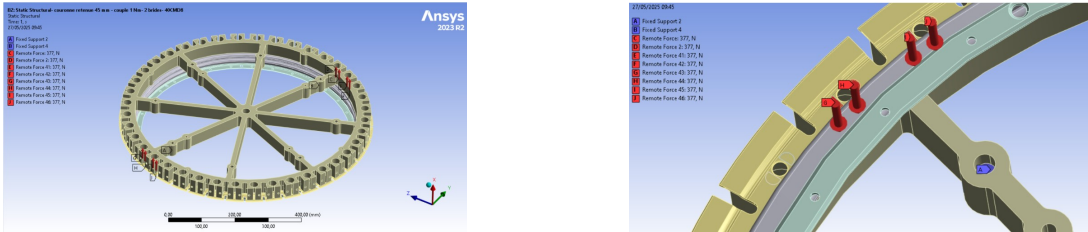
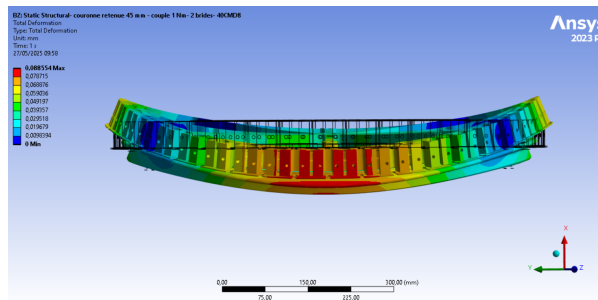
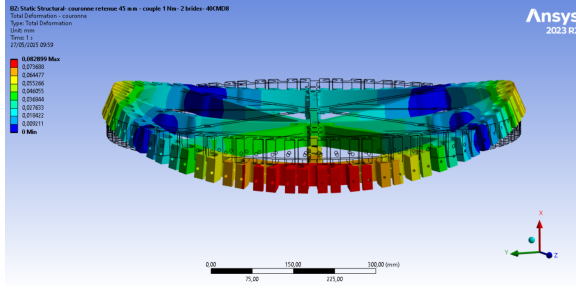


Figure 46: Boundary Conditions - Clamping Torque

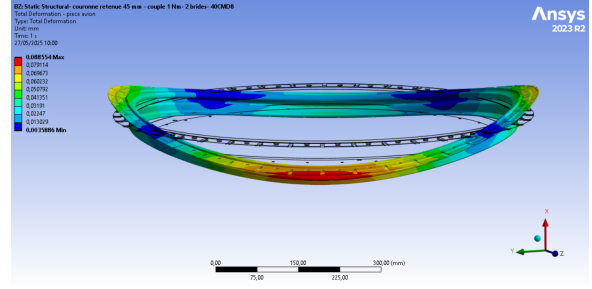
The resulting deformations are presented in Figure 47. To better interpret the mechanical behaviour, the deformations of the crown and sealing flange support were studied separately.



(a) Global deformation



(b) Crown deformation



(c) Support deformation

Figure 47: Deformation - Clamping Torque of 1 Nm

On the amplified scale, a symmetric “U-shape” deformation is once again observed. Separate analysis of the crown reveals a 60% extrapolated deformation ranging between 0.13 and 0.14 mm under a 1 Nm torque.

Table 14 compares the simulated deformation results for the crown under both torque levels.

Table 14: Clamping Torque 1 Nm vs 3 Nm

Torque	Extrapolated Deformation
1 Nm	0.132 mm
3 Nm	0.408 mm

These results confirm the earlier observations: lower clamping torque significantly reduces deformation. A threefold increase in torque results in a proportional increase in deformation, with values rising from 0.132 mm at 1 Nm to 0.408 mm at 3 Nm. This near-linear relationship between torque and deformation clearly supports the selection of the 1 Nm configuration as the most favourable option.

Based on the findings from both simulations, the 45 mm thick crown has been definitively validated for implementation. This decision formally concludes the design phase.

In addressing the third research question (how to ensure manufacturability of the selected solution) this section demonstrated that the final design not only meets performance objectives, but also remains compatible with industrial constraints such as machining, ergonomics, and safety regulations.

In summary, the improved designs fully integrate technical, ergonomic, and financial constraints. These validated refinements establish a solid foundation for the Control phase, which will assess their real-world performance and ensure compliance with the initial requirements. This concludes the Improve stage. The following section will evaluate the effectiveness of the implemented solutions and confirm their suitability against the criteria defined in the early phases of the DMAIC framework.

4.6 Control

This final phase consists in evaluating the performance of the two proposed solutions. The short-term solution constitutes the first stage of response to the problem. Its effectiveness will determine whether it is necessary to proceed with the production of the reinforced crown (long-term solution). Indeed, if the spring height adjustment system proves sufficiently effective, the new crown may not be required to meet the client's flatness tolerance.

4.6.1 Spring Height Adjustment System

As a reminder, the spring height adjustment system includes three inseparable components: the addition of adjustment shims, a cross-tightening sequence applied to the 20 peripheral clamps, and the use of a reduced tightening torque of 1 Nm. The shims theoretically allow precise height adjustment of each spring supporting the crown. This system aims to eliminate the previously measured height variation of ± 0.1 mm between springs with the current configuration.

The evaluation criteria are as follows: the overall functionality of the system, its effectiveness, and the ergonomics of its implementation. The impact on assembly time is also taken into account (does the adjustment introduce significant delays? If so, can it be anticipated or standardised?). The objective is to determine whether this adjustment, once properly calibrated, can ensure dimensional compliance without requiring a new reinforced crown.

Table 15 presents the measured heights of the eight springs when applying the short-term solution (shimming system). Two sets of data are provided: one using a 0.3 mm shim foils and the other a 0.15 mm shim foils. The bottom of the table summarises the average height, minimum and maximum values, and the overall range.

Table 15: Spring Heights - Shim Foils

Spring	0.3 mm	0.15 mm
1	39.98	39.80
2	40.08	39.82
3	40.05	39.79
4	39.98	39.81
5	39.98	40.01
6	40.01	40.16
7	39.89	39.99
8	39.82	39.79
Average	39.97	39.90
Min	39.82	39.79
Max	40.08	40.16
Range	0.26	0.37

Despite various adjustment attempts using 0.3 mm and 0.15 mm shim foils, the measured values shown in remain unsatisfactory. Indeed, the differences between support points are too significant, and the obtained heights deviate substantially from the theoretical value of 39.57 mm. The unstable behaviour of the crown and excessive measurement dispersion make it difficult to achieve a reliable and repeatable adjustment (range between 0.26 and 0.37 mm). As a result, the short-term solution does not ensure dimensional conformity. The fabrication of a new crown was therefore unanimously decided during a follow-up team meeting.

4.6.2 New Self-Supporting Crown

The new self-supporting crown constitutes the long-term solution of the project. At the time of writing, it has not yet been manufactured. However, its performance has been rigorously evaluated through simulations, enabling conclusions to be drawn regarding key aspects such as the design, performance indicators, and compliance with the requirements defined at the beginning of the study.

Design

As shown in the 3D models in Figure 48, the new crown incorporates both thick and thin reinforcements, along with an optimised geometry. The eight original “ears” have been removed in order to maximise stiffness while limiting the increase in total mass. As previously, the current crown appears in turquoise blue, while the new version is shown in purple, making differences and similarities easy to visualise.

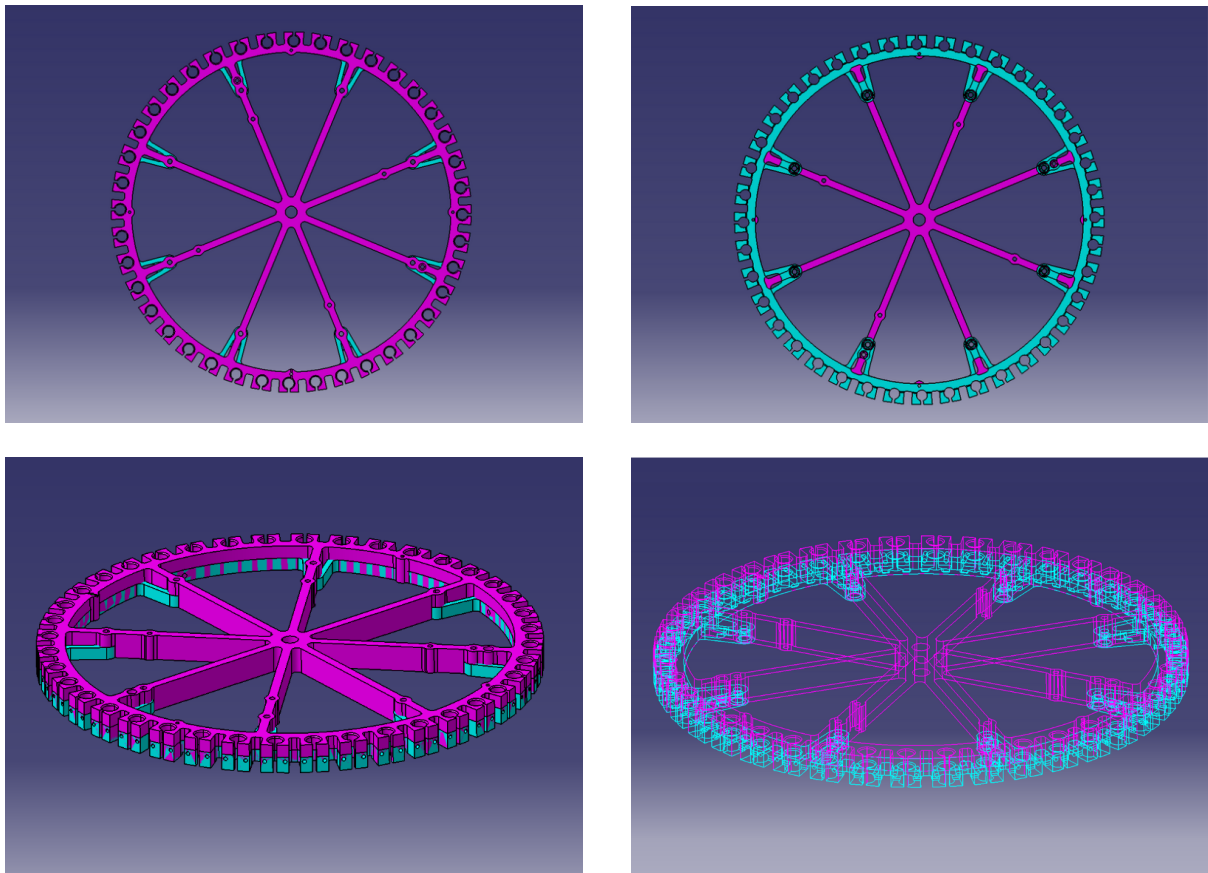


Figure 48: Old Crown vs New Crown - Design

Performance Indicators

Table 16 compares the results obtained for the new crown with those of the current one. A 3-point rating system is used to assess how well each predefined performance indicator has been met.

Table 16: Performance Indicator - New Crown

Indicator	Results	Rating
Stiffness	Theoretical extrapolated deformation (60%) of 0.046 mm vs. 0.25 mm initially (81.6% reduction)	Requirement fully met - 3/3
Mass	30.2 kg (maximum allowable mass) vs 11.37 kg for the current crown	Requirement met with compromise on mass/deformation - 2/3

Although the mass constraint remains stringent, both performance indicators exceed 2/3, validating the new crown. While the theoretical flatness is met, the crown is not deformation-free. It can accommodate minor deviations, which must be clearly specified, but cannot correct major geometric defects. Achieving such rigidity would require a heavy tool comparable to an anvil, conflicting with the project’s goals of ergonomics, handling, and optimised design.

Requirement Compliance

It is also essential to verify compliance with the four major requirements defined in the Methods chapter. This validation confirms whether the new crown meets the initial objectives in terms of mass, geometry, robustness, and environment. Table 17 provides examples supporting the fulfilment of each requirement.

Table 17: Requirement Compliance - New Crown

Requirement	Met	Justification
Mass	Yes	Compliant as explained in the performance table. To be verified under real conditions with the rotary platform. A suitable lifting system must be provided, as the mass exceeds 25 kg (AFNOR standard NF X35-109).
Geometry	Yes	Simple design ensuring accessibility to all 160 probing points and ensuring a theoretical flatness of the controlled part below 0.025 mm. Free height is preserved. Experimental validation is pending.
Robustness	Yes	Selected material: tool steel 40CMD8. Reinforced geometry significantly improves the overall stiffness.
Environment	Yes	Retention of the 8 springs is ensured via counterbored holes despite the removal of the original “ears”. Theoretical compliance with the sealing flange support geometry (to be confirmed in practice).

All requirements are met, confirming the compliance of the new crown. This includes alignment with LAUAK's CSR policy and the client SAFRAN's technical expectations. Furthermore, the design adheres to relevant regulatory standards, including NF X35-109, which governs the manual handling of heavy loads and the associated ergonomic provisions.

Lastly, technical feedback from workshop operators was positive following the delivery of the CATIA 3D model and simulation report. It confirmed that the 160 probing points remain accessible. The toolpath of the probe head will require a slight program modification, but this is not a blocking issue. Regarding equipment, the rotary platform supports the 30 kg load, and one of the available probes has a sufficient stroke (over 45 mm). As a result, the new crown is fully validated.

In conclusion, the Control phase confirms that the selected solution complies with the technical, ergonomic, and industrial requirements defined at the beginning of the study. The spring adjustment system, although initially promising, failed to meet the dimensional conformity criteria in a reliable and repeatable manner due to the insufficient rigidity of the current crown. Consequently, the development of a new reinforced crown was deemed necessary.

The optimised crown design was validated both through numerical simulations and by technical stakeholders. It offers a substantial reduction in deformation, ensures accessibility for inspection, and remains compatible with existing equipment. Moreover, it satisfies all predefined performance indicators and design constraints, including mass, geometry, robustness, and environmental integration.

Despite the robustness of the approach and the consistency of the results obtained, certain limitations must be acknowledged. The simulations were conducted under assumptions of linear material behaviour and idealised loading conditions, which may slightly differ from actual industrial scenarios. In addition, reinforcement configurations were assessed individually, without combined evaluations or tolerance analyses, which somewhat limits the generalisability of the findings. These considerations open up opportunities for future investigations that are more refined and better aligned with real-world operating conditions.

This final validation closes the DMAIC improvement cycle, providing LAUAK with a robust and operational dimensional control solution adapted to the constraints of the LEAP-B1 sealing flange support.

5 Conclusion

The increasing demand for high-precision dimensional control in the aerospace industry, particularly for lightweight and flexible components, has highlighted the limitations of conventional inspection fixtures. The studied case, an inspection tool for the sealing flange support of the LEAP-B1 engine, revealed critical challenges related to repeatability and deformation, with measured flatness deviations far exceeding the specified 0.025 mm tolerance.

This study aimed to redesign the existing self-supporting crown system to ensure both dimensional accuracy and industrial feasibility. The work followed a rigorous and structured methodology based on the DMAIC approach, integrating a multi-criteria analysis, experimental measurements, CATIA parametric modeling, and finite element simulations in ANSYS.

Several reinforcement configurations were explored, including internal stiffeners, extra thickness, and hybrid solutions, while maintaining compatibility with the part's geometry and operational constraints. The simulations showed that certain reinforced designs, especially those involving internal stiffeners or combined strategies, significantly reduced the deformation (down to values compliant with the tolerance) without exceeding mass or ergonomics constraints.

The principal achievement of this research is the development and digital validation of an improved inspection fixture capable of meeting the geometric requirements set by SAFRAN. The proposed solution remains compatible with LAUAK's CSR objectives, ensuring a reliable and industrially viable alternative to the current tool. It also demonstrates that digital simulation can replace costly trial-and-error prototyping phases.

Nonetheless, the study is subject to several limitations. The proposed design has not yet undergone physical prototyping or experimental validation. Moreover, some phenomena such as thermal effects, material aging, and operator variability were not included in the simulations.

To extend this work, the following developments are recommended:

- Prototype the redesigned fixture and compare simulated versus experimental deformations.
- Investigate advanced materials to further reduce deformation while maintaining low weight.
- Conduct an experimental sensitivity analysis including thermal and operator-induced variables.
- Explore potential integration of sensors for real-time monitoring during inspection.

Despite these limitations, this study successfully fulfilled its initial objectives by proposing a validated and realistic design improvement. It showcases a methodologically sound and application-oriented process to address tooling issues in industrial metrology.

The implications of this work extend beyond the specific case study. The proposed methodology can be adapted to other flexible aerospace components where dimensional accuracy is critical. Moreover, the integration of CAD and FEA in early development stages paves the way for more robust and cost-efficient inspection solutions in the long term.

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A Appendices



10 June 2025

Dear Mr Mathieu Doyenard ,

Reference: CURES/25611/2025

Project ID: 29040

Title: Improvement of the Three-Dimensional Inspection Tooling

Thank you for your application to the CURES system.

We are pleased to inform you your CURES application, reference CURES/25611/2025 has been reviewed. You may now proceed with the research activities you have sought approval for.

If you have any queries, please contact CURES Support.

We wish you every success with your project.

Regards,

CURES Team

Reference	Summary / Main Argument	Methodology	Findings / Conclusions	Themes	Critical Appraisal (Limitations)
AFNOR NF EN ISO 1101, Spécification géométrique des produits (GPS) - Tolérancement géométrique, 2006, Available at: https://jackadit.com/cours/metro/ISO1101-janvier2006.pdf	Standardized rules for geometric dimensioning and tolerancing in industrial design and manufacturing	International standard document	Defines consistent tolerancing symbols and values to ensure form, orientation, and location specifications across industries	Dimensional standards Geometric tolerancing Technical drawings	Highly formal and descriptive / Lacks practical case studies or direct application examples for fixturing
ZEISS – Quality Forum, ISO 10360 - The New Acceptance and Reverification Tests for Coordinate Measuring Machines. Available at: https://qualityforum.zeiss.com/migration/images/120_2230526f9bdb0832c207ec0e65c8986e.pdf	High-precision industrial measurement methods Importance of dimensional control in manufacturing	Technical presentation / Industrial guide	Dimensional precision depends on correct measurement strategies and appropriate fixturing	Dimensional control Measurement methods	Limited details on fixture / workpiece interaction under low-load conditions
Metrology.news, Differentiating Between Tactile and Optical Measurement Technology for Dimensional Inspection, 2024, Available at: https://metrology.news/differentiating-between-tactile-and-optical-measurement-technology-for-dimensional-inspection/	Comparison of tactile and optical dimensional inspection technologies in terms of speed, accuracy, and suitability	Industrial article	Tactile methods are precise but slow and invasive / Optical methods are faster and non-contact but less suitable for some surfaces	Measurement methods Metrology technologies	Lacks quantitative analysis or detailed metrological data
CarrLane, Fixture Design Principles. Available at: https://www.carrlane.com/engineering-resources/fixture-design-principles/locating-clamping-principles	Principles of locating and clamping without constraint / Design of precision measurement fixtures	Industrial engineering technical guide	Poorly designed clamping can induce deformations / Importance of constraint-free locating	Fixture design Constraint-free positioning	General focus / Not specific to ultra-low deformation requirements (<0.025 mm)
Shi Y. et al., Design and Manufacture of a Flexible Adaptive Fixture for Precision Grinding of Thin-Walled Bearing Rings, 2025, Available at: https://www.mdpi.com/2504-4494/9/5/139?utm_source=chatgpt.com	Proposes a fixture design adaptable to thin-walled components to minimize deformation during precision grinding	Research article with finite element and modal analysis	Integrating flexible clamping features improves part integrity during inspection by validated through FEM	Adaptive fixturing Fixture design FEM	Focused on grinding processes / Requires generalization for broader dimensional inspection contexts
St. Marys Spring Company, Precision Mechanical Springs for Critical Applications, Available at: https://www.stmarysspring.com/precision-mechanical-springs/	Spring design factors affecting mechanical precision and reliability in support and clamping applications	Technical article from industry supplier	Spring performance is governed by geometry and material selection, directly affecting fixture compliance and	Spring systems Compliance control Adjustable fixturing	Useful mainly as design guidance not for self-supporting fixture
Lin S. et al, Structure Design Improvement and Stiffness Reinforcement of a Machine Tool through Topology Optimization Based on Machining Characteristics, 2024. Available at: https://www.mdpi.com/2076-3417/14/1/61	Design of adjustable fixtures for inspection of complex workpieces / Optimization of rigidity	Research article with case studies and modeling	Adjustable locating devices improve measurement precision by reducing residual stresses	Rigidity optimization Adjustable fixtures	Does not cover effects of aging or thermal loads on fixtures
Dr Kovacevic A., CATIA V5, University of London, Available at: https://www.staff.city.ac.uk/ra600/Presentations/CATIAV5Lectures.pdf	CATIA V5 environment and features	Lecture material from university course	CATIA V5 supports integrated modeling, assembly, and drafting for 3D inspection fixture development	CAD design Software Engineering	Focused on software usage (education) / Not specific to metrology tooling

Appendix 2: Source Review Grid (1/2)

Reference	Summary / Main Argument	Methodology	Findings / Conclusions	Themes	Critical Appraisal (Limitations)
Wang S. et al, Parametric Design and Associativity in 3D CAD, 2008. Available at: https://www.researchgate.net/publication/250349102_Parametric_Design_and_Design_Associability_in_3D_CAD	Importance of parametric design in CAD for flexibility and assembly associativity	Research article / Review on CAD modeling	Parametric design facilitates updates across assemblies and improves model quality	Parametric design Industrial CAD	No direct application to inspection fixtures / General CAD framework
Lee H., Finite Element Simulation with ANSYS workbench 2019, 2019. Available at: https://books.google.fr/books?hl=en&lr=&id=QPqcDwAAQBAJ&oi=fnd&pg=PP3&dq=Finite+element+analysis+of+mechanical+parts+under+loading+and+boundary+conditions+using+ANSYS+Workbench&ots=ST2KodOkYD&sig=w_T0Cca2O1UIQjDFBw6LsiHYd8&redir_esc=y#v=onepage&q&f=false	Use of ANSYS Workbench for validation of stresses, deformations, and flatness under load	Technical book on FEM simulations	FEM allows prediction of critical mechanical deformations before manufacturing	FEM simulation Mechanical validation	No explicit treatment of spring-supported boundary conditions
Hemesh P. et al., Mesh convergence study and estimation of discretization error of hub in clutch disc with integration of ANSYS, Available at: https://iopscience.iop.org/article/10.1088/1757-899X/402/1/012065/pdf	Mesh refinement in FEM to reduce discretization error and ensure stable simulations	Research article on FEM simulation with convergence study	Progressive mesh refinement leads to convergence of stress and displacement, essential for result credibility but increase calculation time	FEM simulation Mesh convergence	Does not address soft support modeling or fixturing constraints
Roberts G. et al, Tooling Steel, 1998. Available at: https://books.google.fr/books?hl=en&lr=&id=ScphevR_eP8C&oi=fnd&pg=PA1&dq=tooling+steel&ots=PbA_2QpBbA&sig=esyOM-wn23aNpCzTXDxP1J58t5M&redir_esc=y#v=onepage&q=tooling%20steel&f=false	Mechanical properties of tooling steels (P20/40CMD8 type) / Impact on rigidity and machinability	Technical book on metallurgy	Mechanical properties condition load behavior and dimensional stability	Materials Mechanical properties	Does not specifically address steels for metrology applications
Mantle, Tool steels and injection modeling: a comprehensive guide, 2024, Available at: https://mantle3d.com/blogs/tool-steels-and-injection-molding-a-comprehensive-guide/?utm_source=chatgpt.com	Tool steels characteristics including machinability, thermal properties, and use in precision tooling	Technical blog article	P20 steel (40CMD8) offers excellent machinability and stability, ideal for high-precision fixturing	Tool material Steel properties	Focused on injection molding but transferable to metrology tooling
Weber D. et al., Analysis of machining-induced residual stresses of milled aluminum workpieces, their repeatability, and their resulting distortion, 2021, Available at: https://link.springer.com/article/10.1007/s00170-021-07171-7?utm_source=chatgpt.com#Sec1	Machining operations generate residual stresses in workpieces, which can cause distortion and alter dimensional accuracy	Experimental study using X-ray diffraction	Residual stresses vary depending on process parameters and tool paths	Residual stress Machining	Limited to aluminum alloys and may not fully generalize to tool steels
Soori M. et al., A Review in Machining-Induced Residual Stress, 2022, Available at: https://hal.science/hal-03679993/document	Machining-induced residual stresses are a key factor influencing component distortion, fatigue life, and performance	Journal of New Technology and Materials	Residual stress is influenced by material properties, tool geometry, cutting parameters, cooling methods, and machining strategy	Residual stress Machining	Review-based approach, no new data / Lacks comparative quantitative analysis between mitigation techniques / May not address industry-specific constraints
Chase W. et al., Design of a Variable Stiffness Spring with Human-Selectable Stiffness, 2023, Available at: https://www.researchgate.net/publication/368842466_Design_of_a_Variable_Stiffness_Spring_with_Human-Selectable_Stiffness	Manually adjustable spring system with selectable stiffness levels via discrete anchoring points	Research article with physical testing	Practical stiffness adjustment without part replacement / Useful for adaptable fixturing	Adjustable systems Spring mechanics	Targeted at biomechanics / Adaptation needed for mechanical tooling and metrology use cases

Appendix 2: Source Review Grid (2/2)